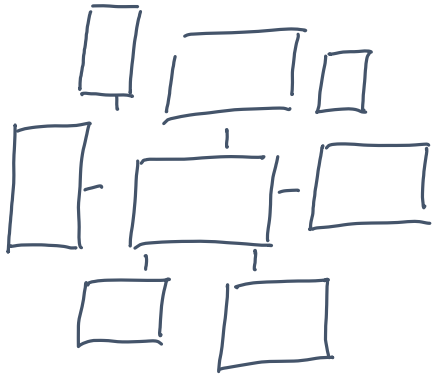




Embedded Signal Processing Systems

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Prof. Dr. C. Jakob

University of Applied Sciences Darmstadt

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Introduction to SystemVerilog HDL design

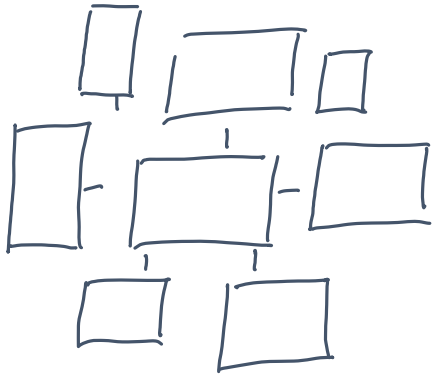
Today's Agenda

Intended topics for today's session

- What is an HDL and how to learn it?
- Learning by example – A first taste: A series of basic combinational and sequential SystemVerilog designs

Appendix

- History of SystemVerilog



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Introduction to SystemVerilog HDL design

Objectives

By the end of this lecture you will be able to ...

- understand the basic structure of a SystemVerilog module
- design simple combinational circuits in SystemVerilog



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Introduction to SystemVerilog HDL design

Recommended Readings

Textbooks, Application Notes, White Papers ...

- Sutherland, S., “RTL Modeling with SystemVerilog for Simulation and Synthesis: Using SystemVerilog for ASIC and FPGA Design”, CreateSpace Independent Publishing Platform, 2017
- Spear, C., “SystemVerilog for Verification: A Guide to Learning the Testbench Language Features”, Springer, 3rd edition, 2012.



FPGAs in a Nutshell – Basic Architectural Features

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FPGAs in a Nutshell ...

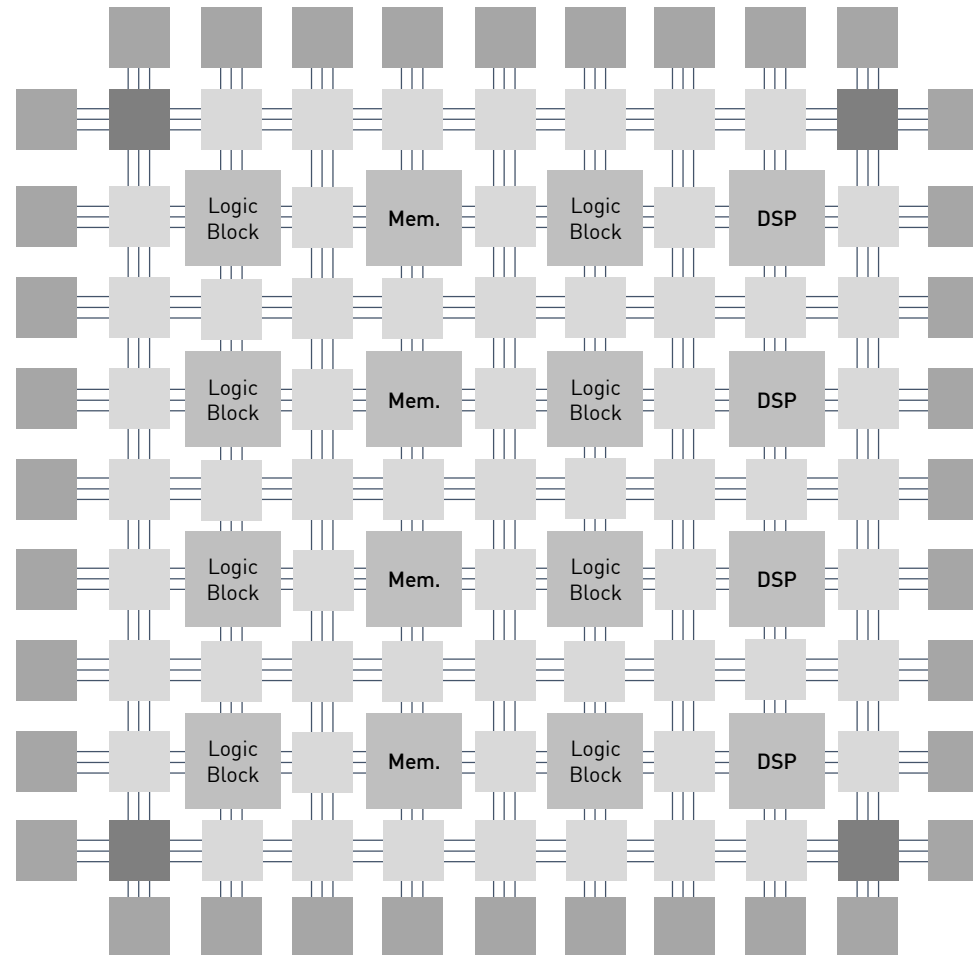
- Everything starts with ...



Intel FPGA

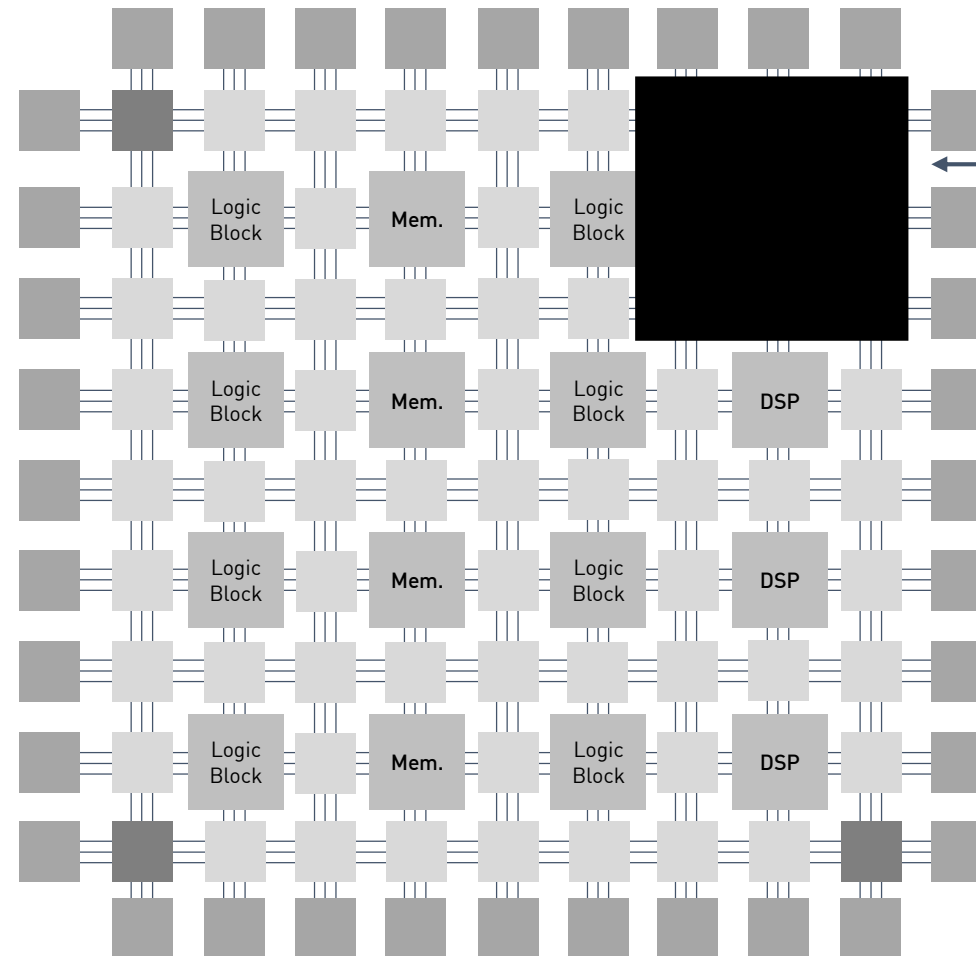
FPGAs in a Nutshell ...

- Simplified FPGA (logic-centric) Structure

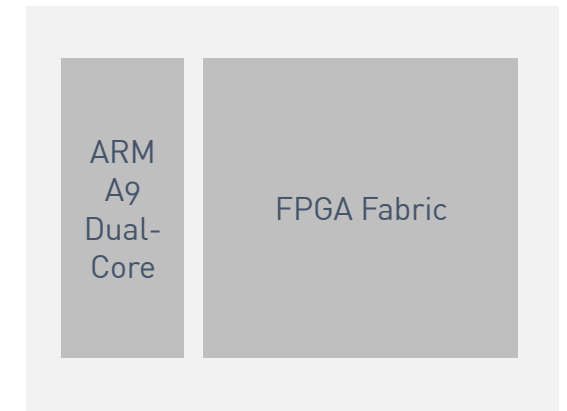


FPGAs in a Nutshell ...

- Simplified FPGA (SoC-centric) Structure



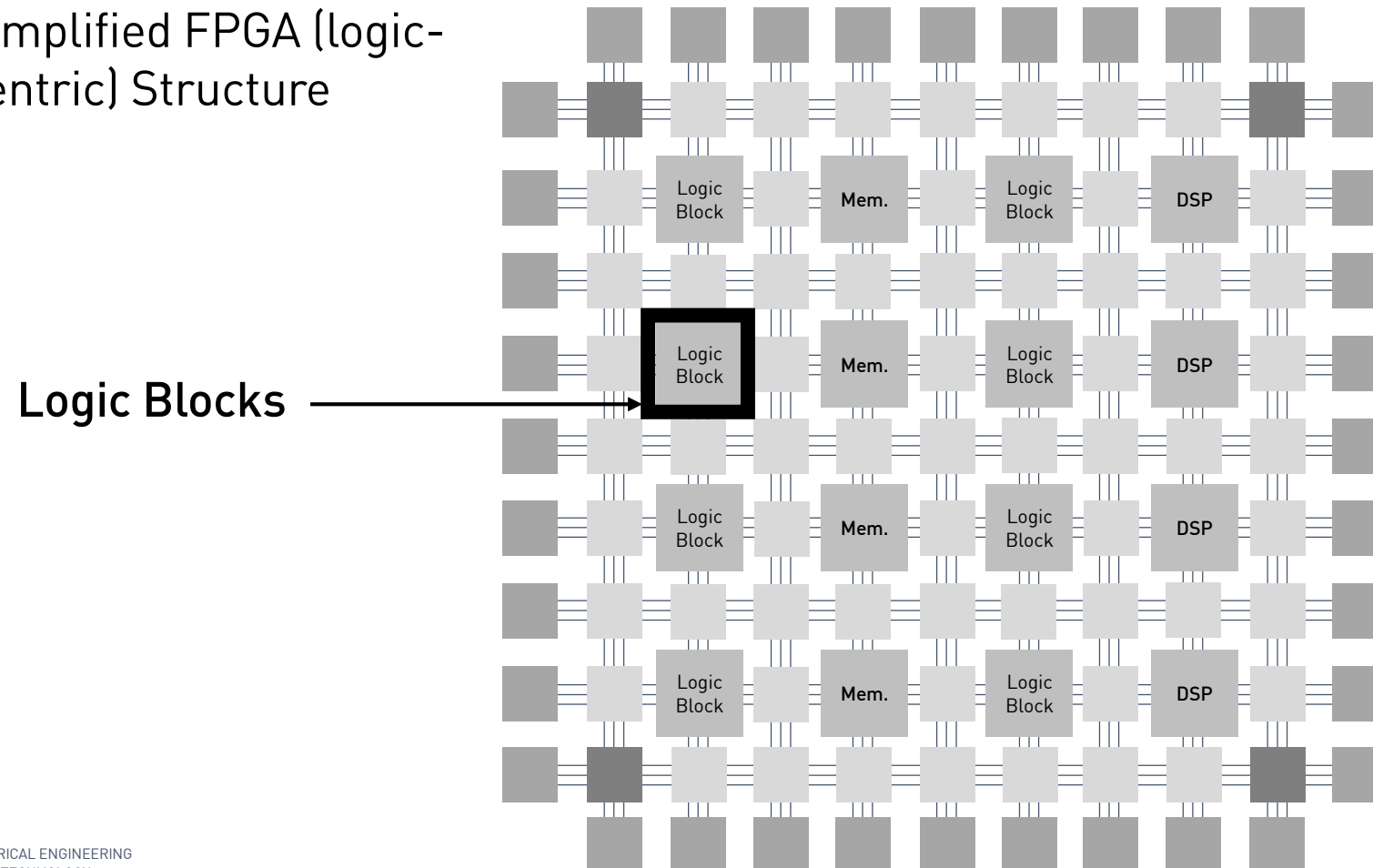
Hard Processor
Systems



Intel SoC FPGA

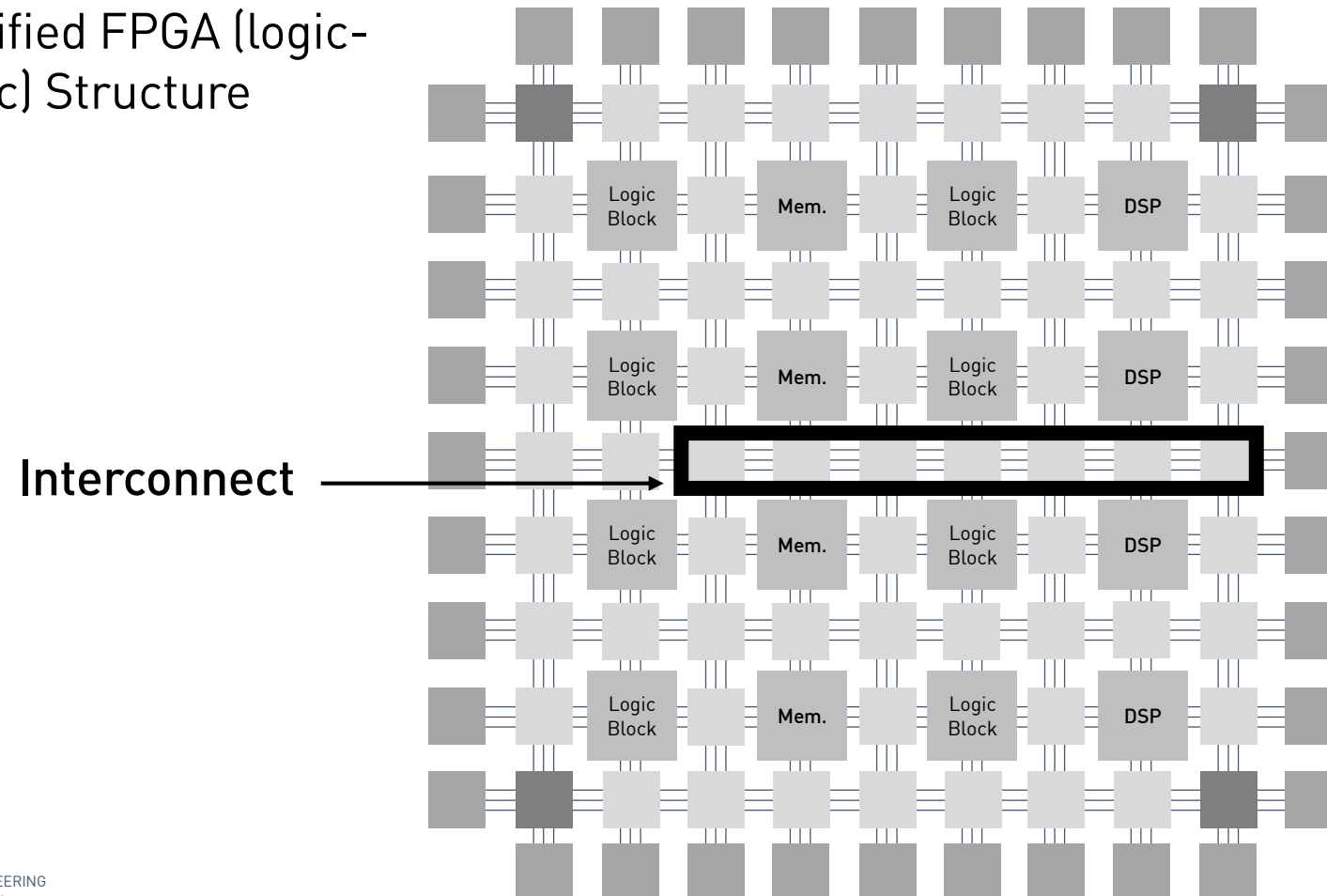
FPGAs in a Nutshell ...

- Simplified FPGA (logic-centric) Structure



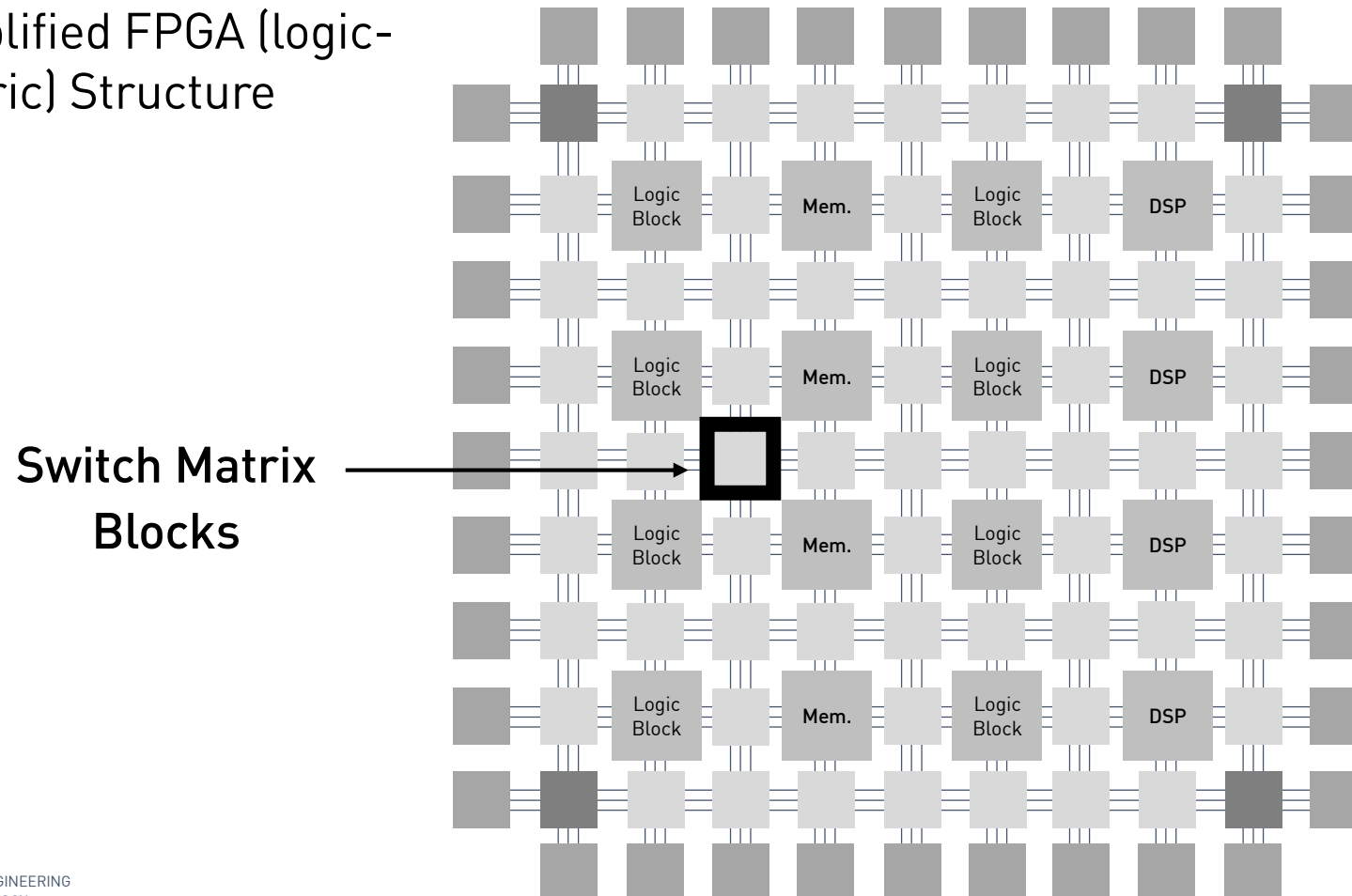
FPGAs in a Nutshell ...

- Simplified FPGA (logic-centric) Structure



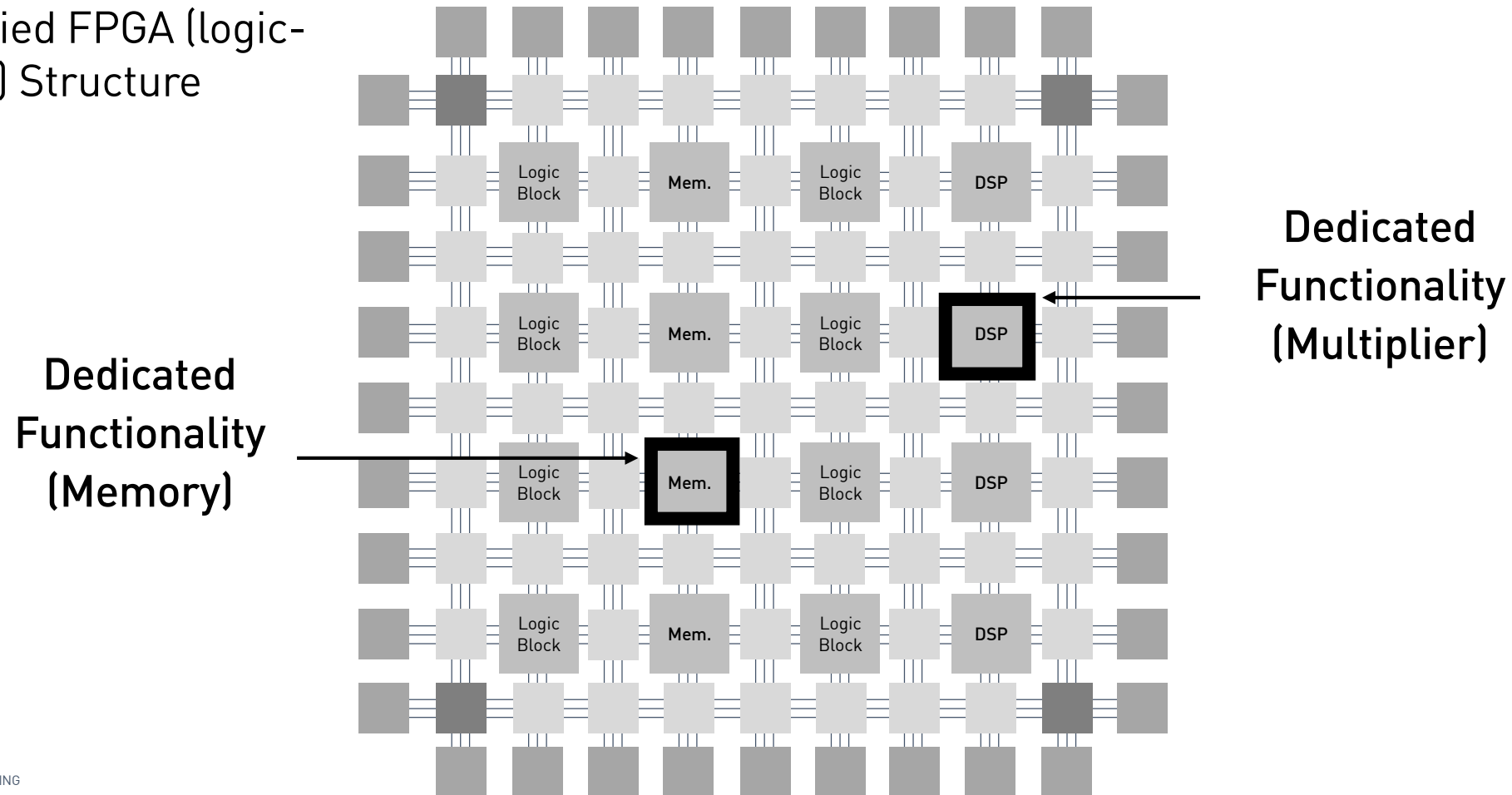
FPGAs in a Nutshell ...

- Simplified FPGA (logic-centric) Structure



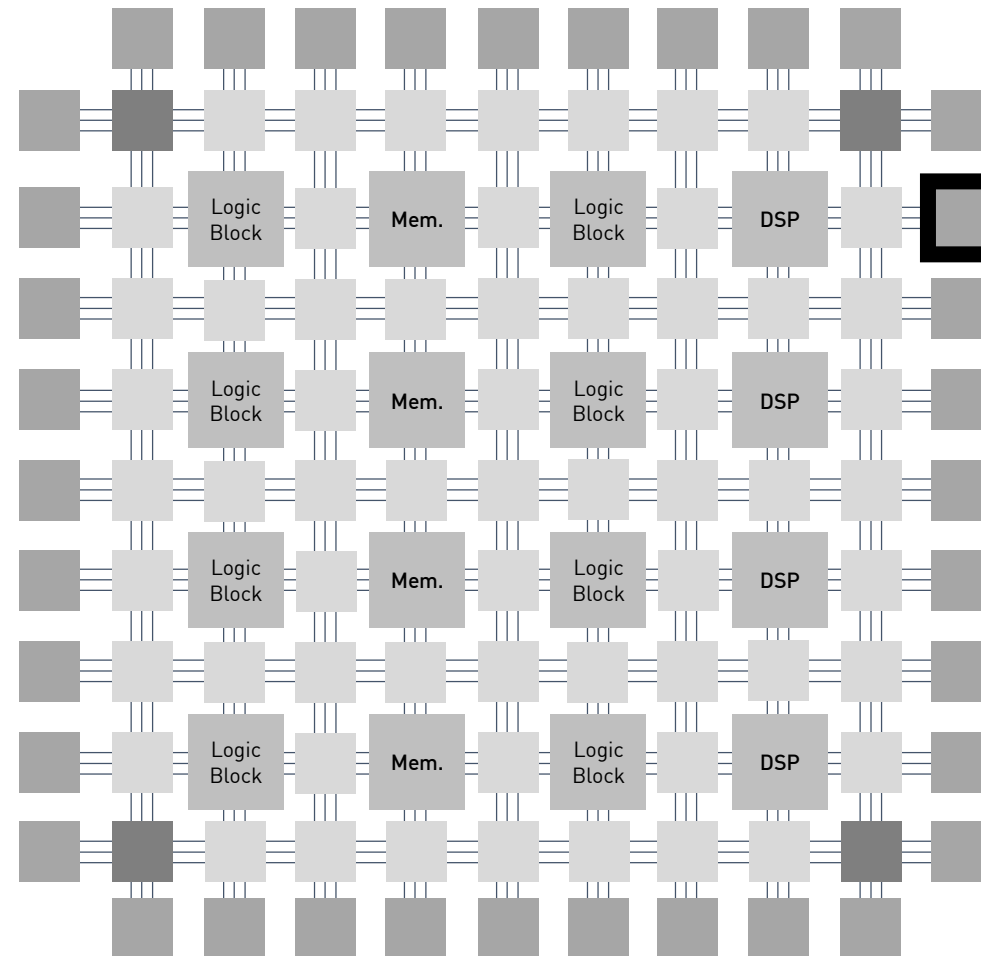
FPGAs in a Nutshell ...

- Simplified FPGA (logic-centric) Structure



FPGAs in a Nutshell ...

- Simplified FPGA (logic-centric) Structure

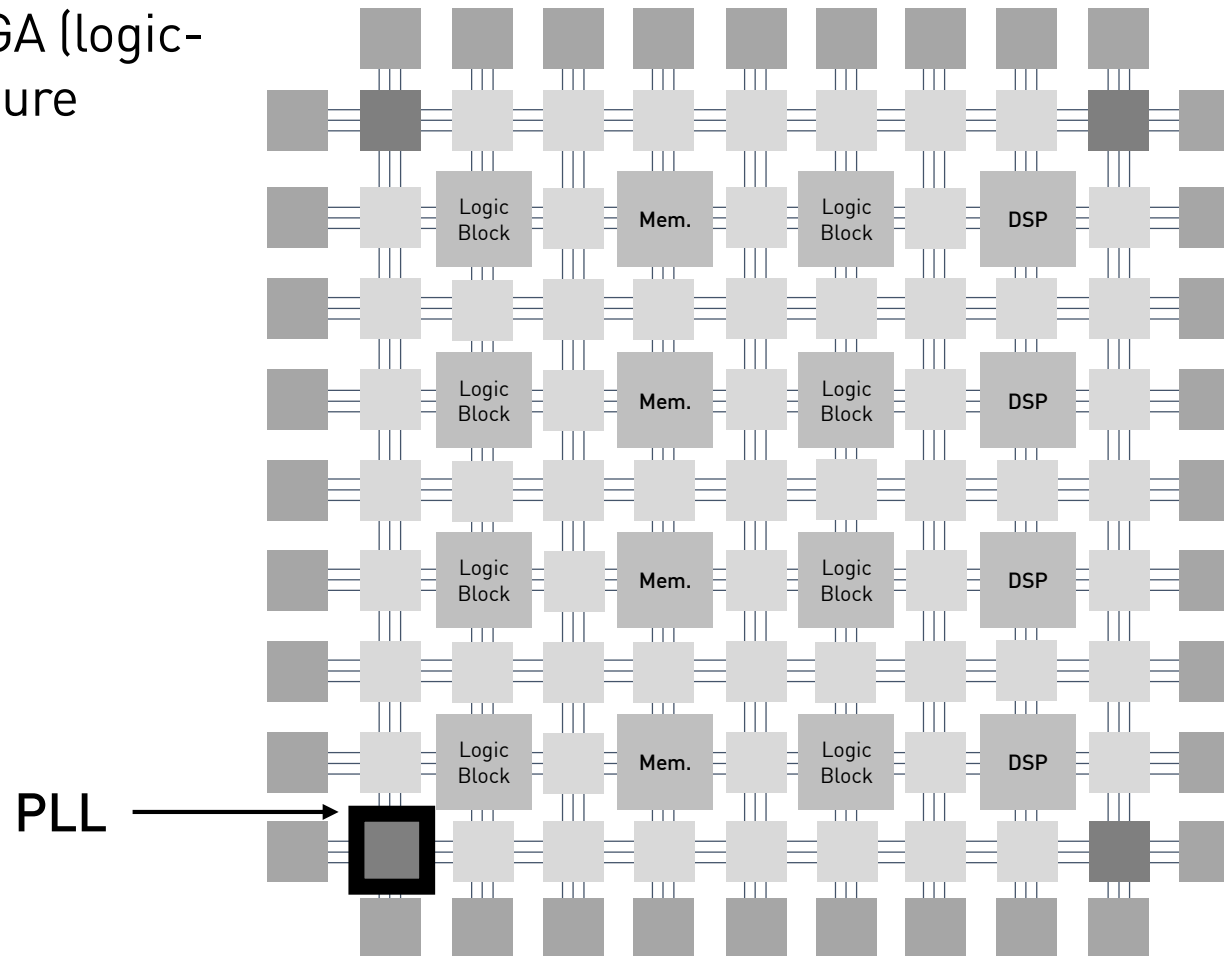


I/O Blocks

- Supporting various I/O standards

FPGAs in a Nutshell ...

- Simplified FPGA (logic-centric) Structure





FPGAs in a Nutshell – Mapping HDL to FPGA logic cells

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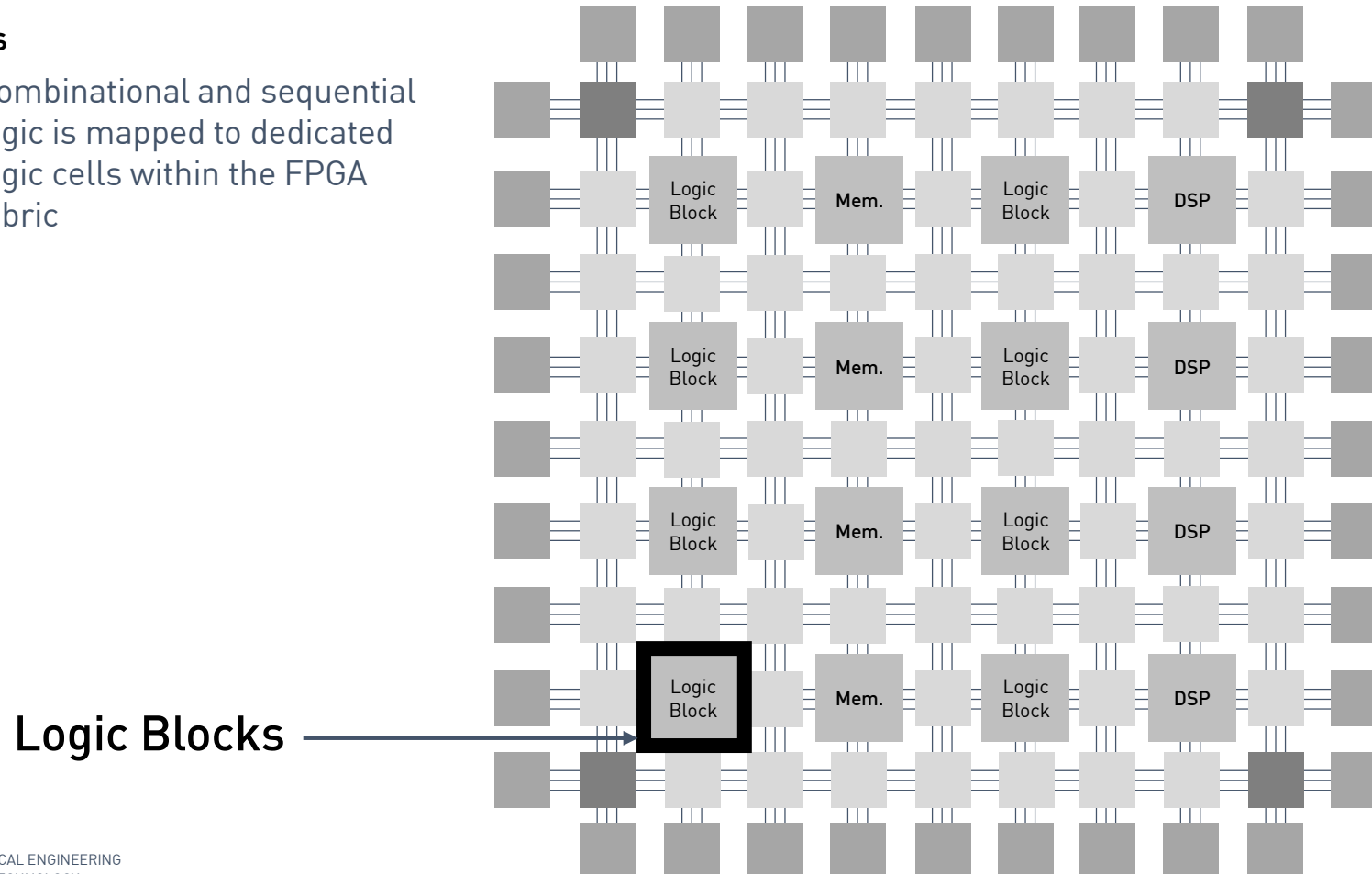
Embedded Signal Processing Systems

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Simplified FPGA Architecture

Notes

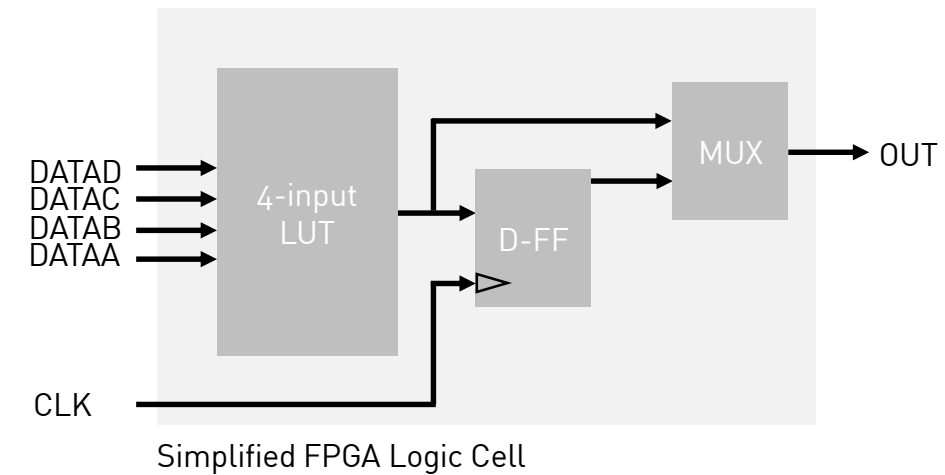
- Combinational and sequential logic is mapped to dedicated logic cells within the FPGA fabric



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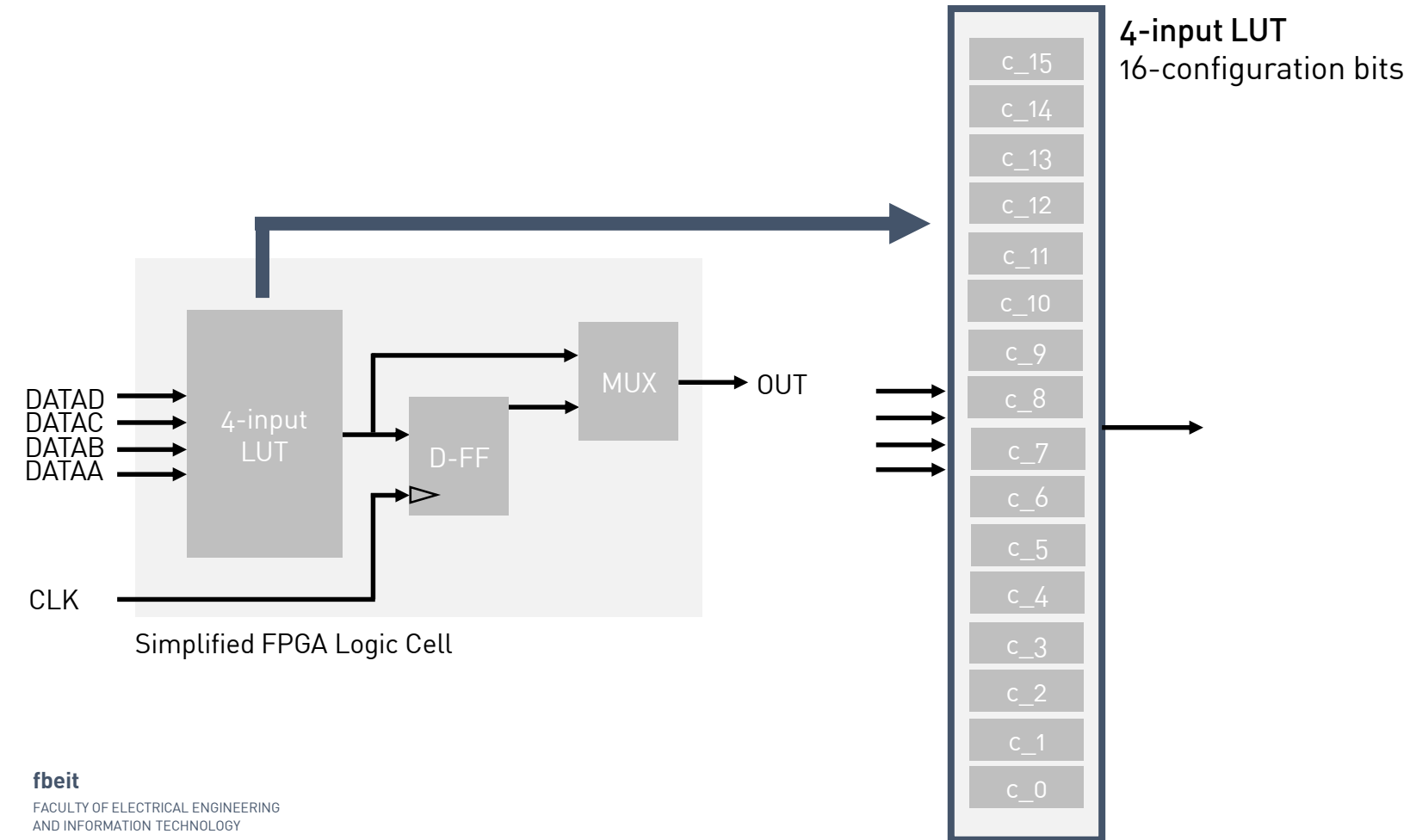
Basic FPGA Logic Cell Structure



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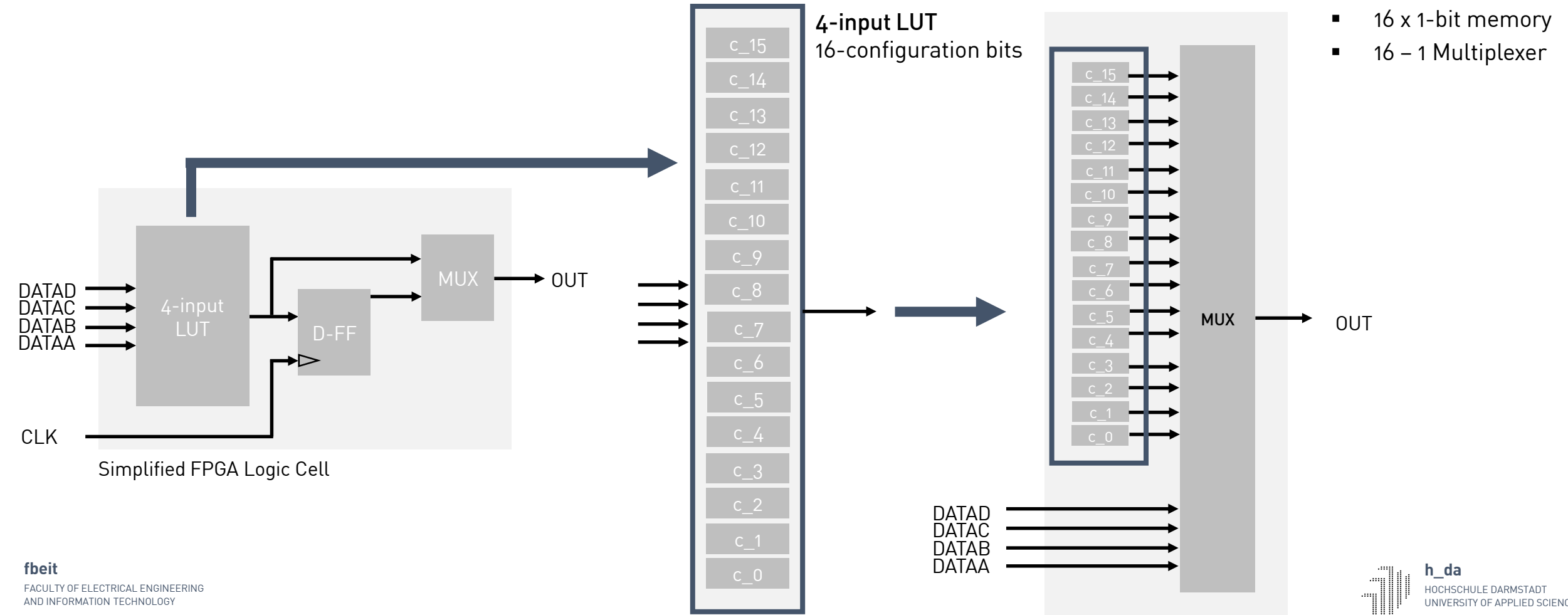
Basic FPGA Logic Cell Structure



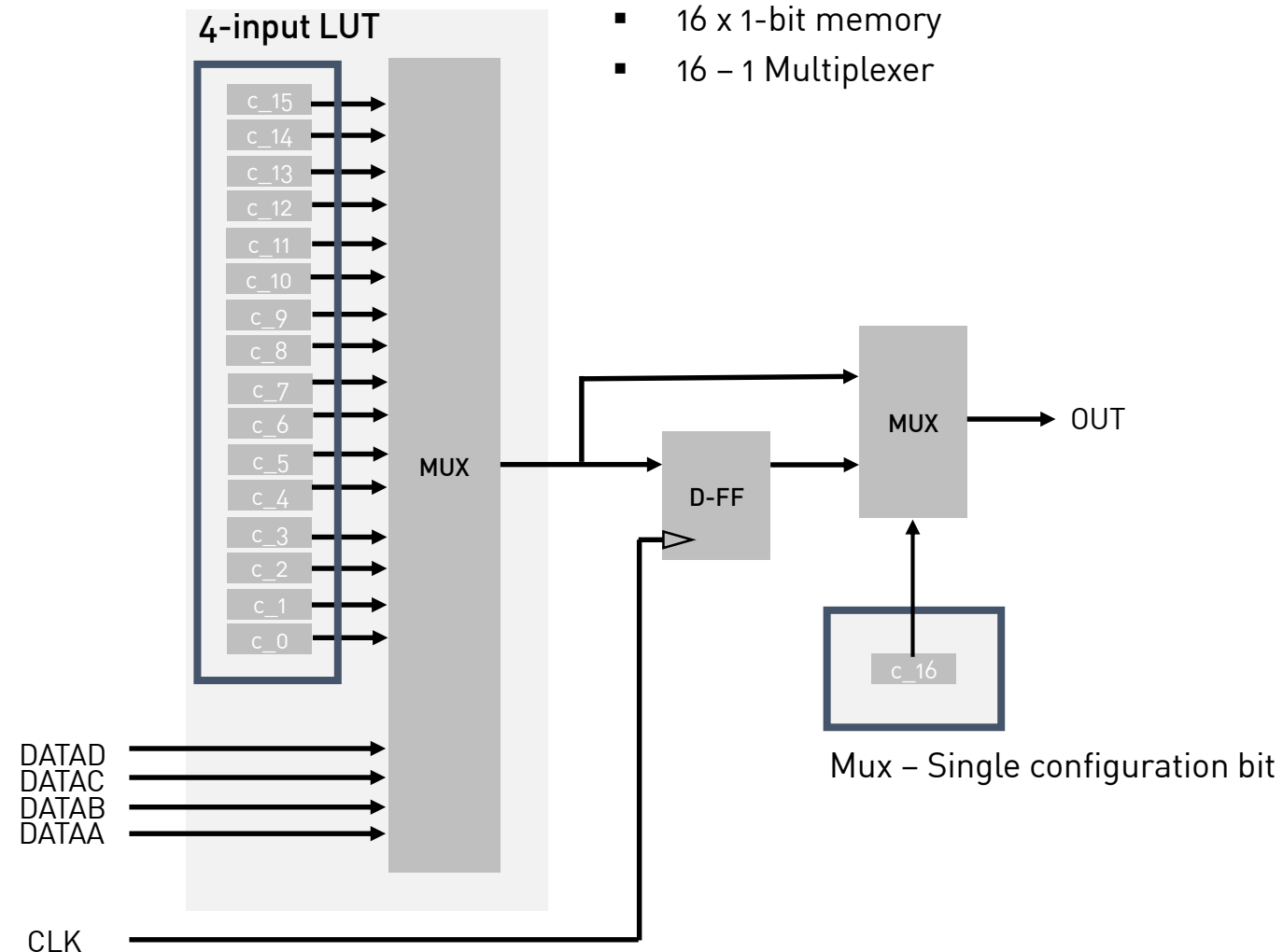
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Basic FPGA Logic Cell Structure



Basic FPGA Logic Cell Structure



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Mapping simple logic functions to the Intel Cyclone IV/X FPGA

Using SystemVerilog

Use case:

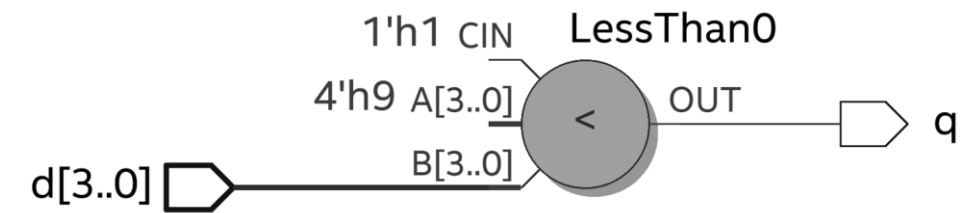
- Logic function implementation: Proof if a 4-bit logic input is greater than or equal to 9

Greater than or equal to 9

```
1. module bool_function_v1(  
2.     input logic [3:0] d,  
3.     output logic q  
4. );  
5. // ### logic function implementation ...  
6. assign q = (d >= 9) ? 1'b1 : 1'b0;  
7.  
8. endmodule
```

1/1

[SystemVerilog] Source Code: bool_function_v1.sv - logic function >= 9

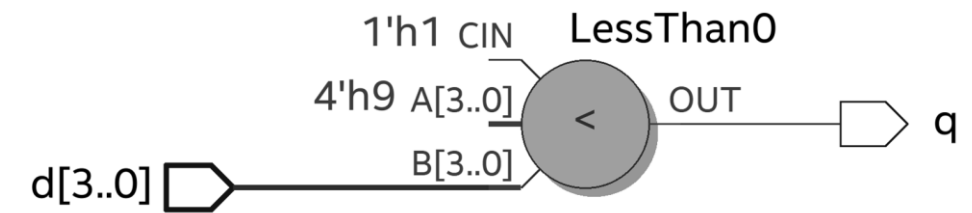


Synthesized circuit

Greater than or equal to 9

```
1. module bool_function_v2(  
2.     input logic [3:0] d,  
3.     output logic q  
4. );  
5. // ### logic function implementation ...  
6. always_comb  
7.     if (d >= 9)  
8.         q = 1'b1;  
9.     else  
10.        q = 1'b0;  
11.  
12. endmodule
```

1/1



Synthesized circuit

[SystemVerilog] Source Code: bool_function_v2.sv - logic function >= 9

Notes

- Leads to the exact same synthesis result ...

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Greater than or equal to 9

```
1. module bool_function_v2(  
2.     input logic [3:0] d,  
3.     output logic q  
4. );  
5. // ### logic function implementation ...  
6. always_comb  
7.     if(d >= 9)  
8.         q = 1'b1;  
9.     else  
10.        q = 1'b0;  
11.  
12. endmodule
```

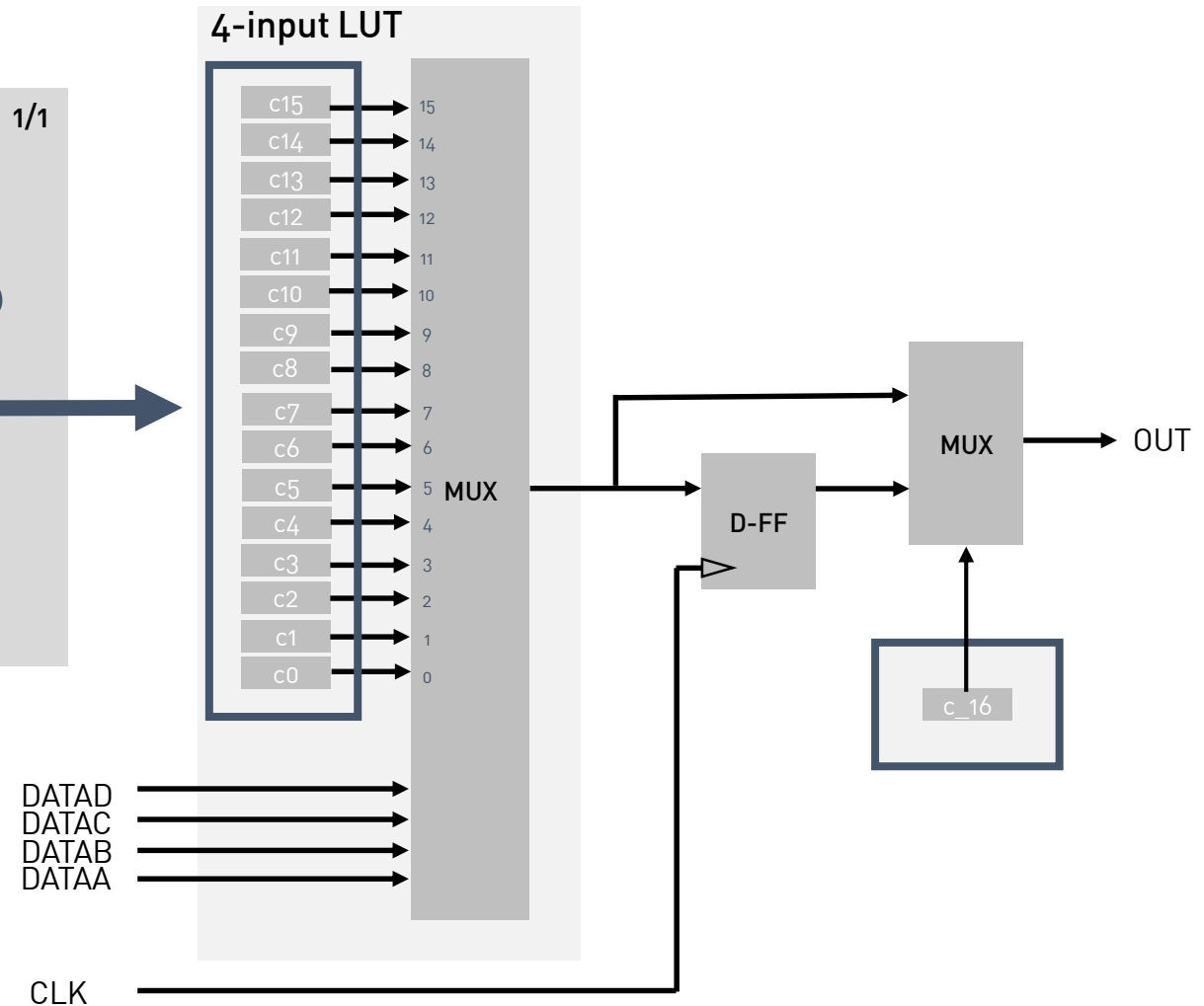
1/1

?

[SystemVerilog] Source Code: bool_function_v2.sv - logic function ≥ 9

Notes

- Leads to the exact same synthesis result ...



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Greater than or equal to 9

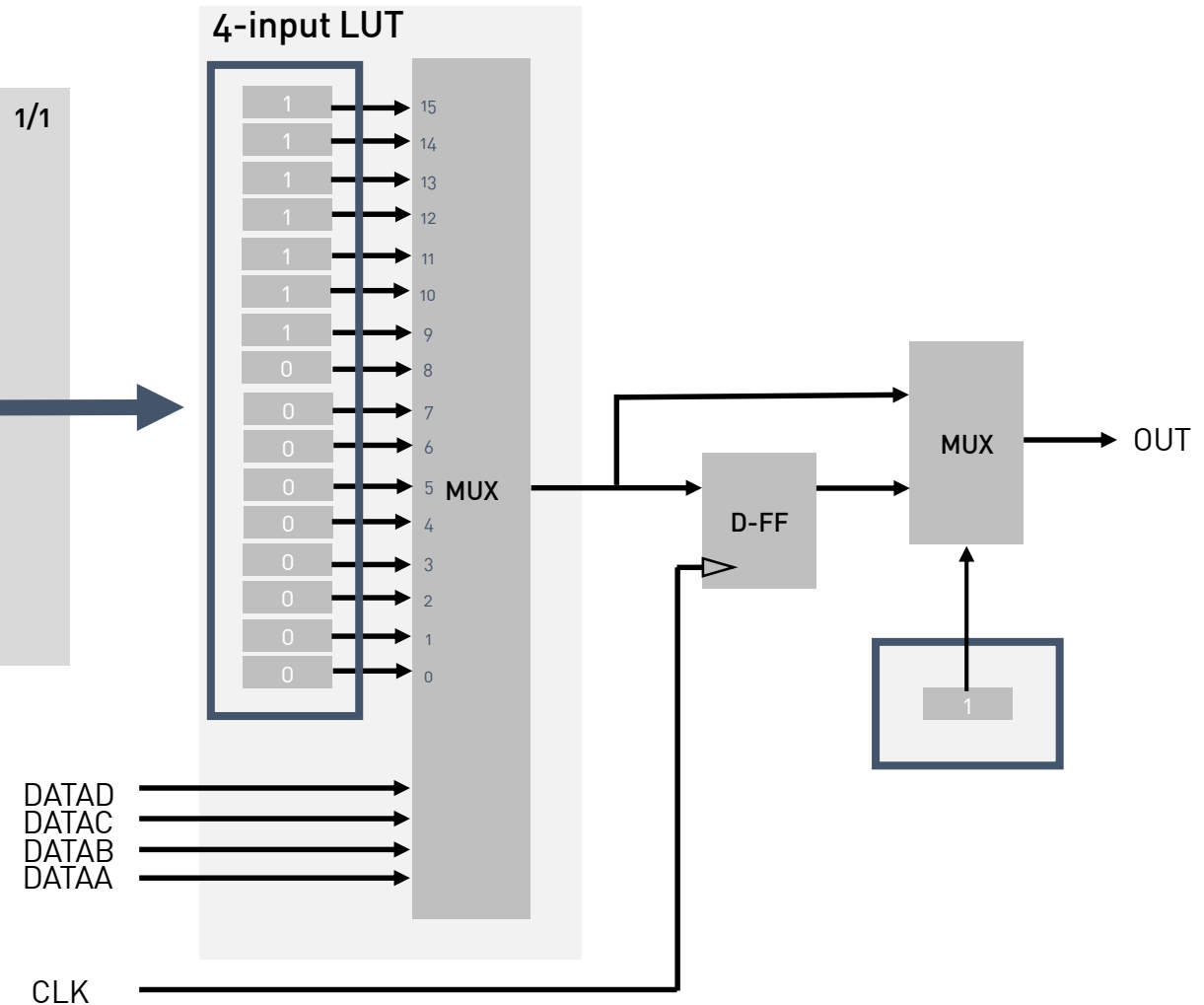
```
1. module bool_function_v2(  
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11.  
12. endmodule
```

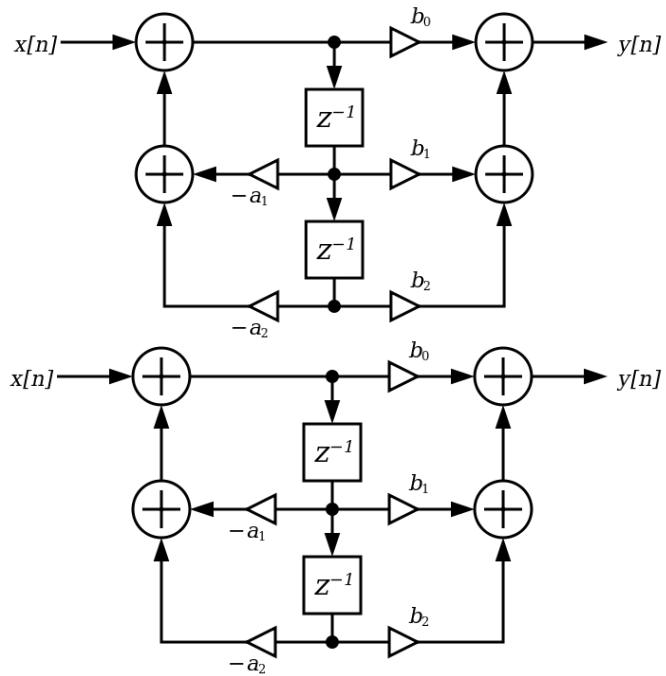
1/1

[SystemVerilog] Source Code: bool_function_v2.sv - logic function ≥ 9

Notes

- Leads to the exact same synthesis result ...





FPGAs in a Nutshell – FPGA Embedded Softcore Processors

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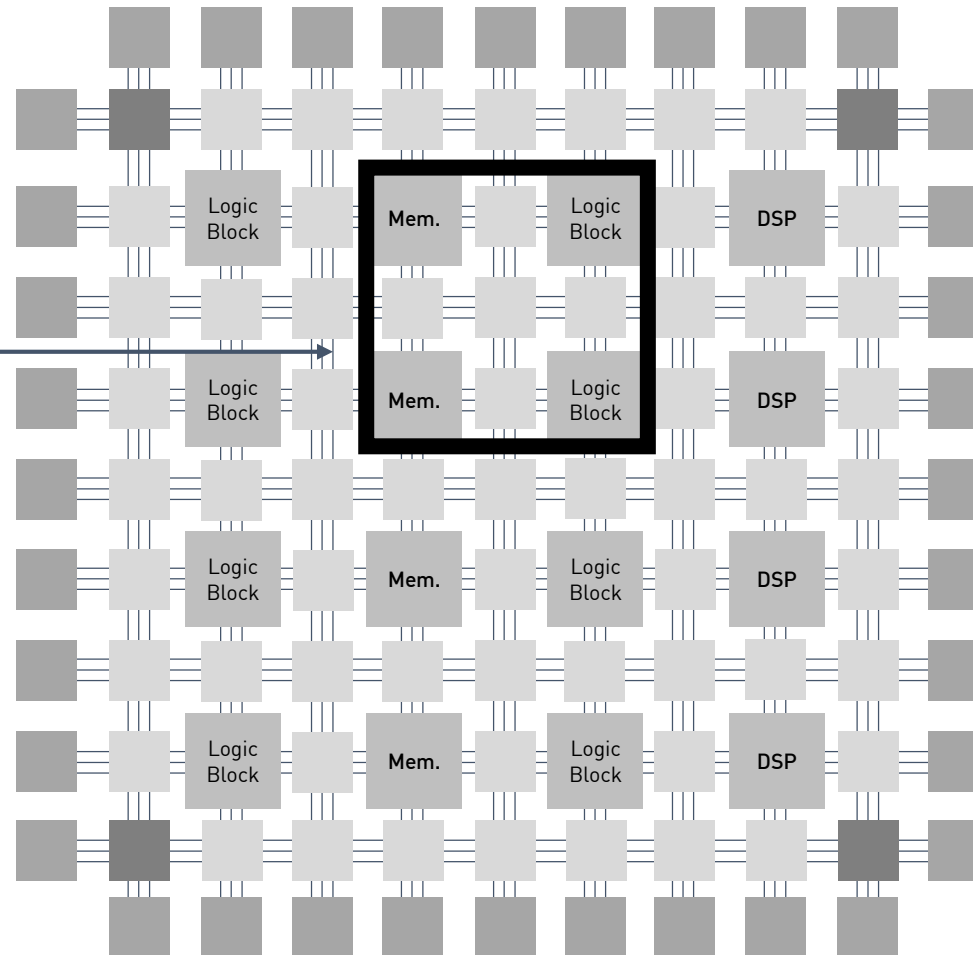
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FPGAs in a Nutshell ...

- Simplified FPGA (logic-centric) Structure

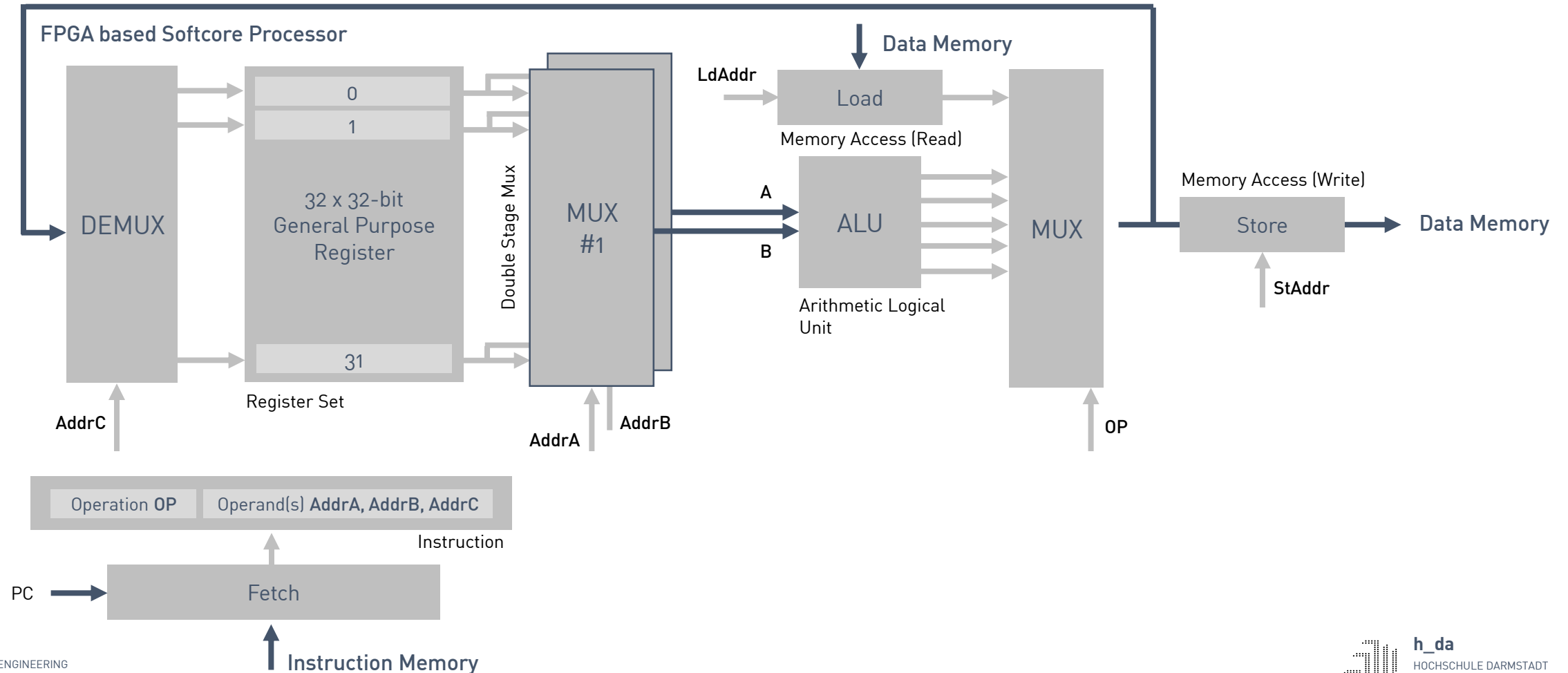
Softcore
Processor



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FPGAs in a Nutshell ...



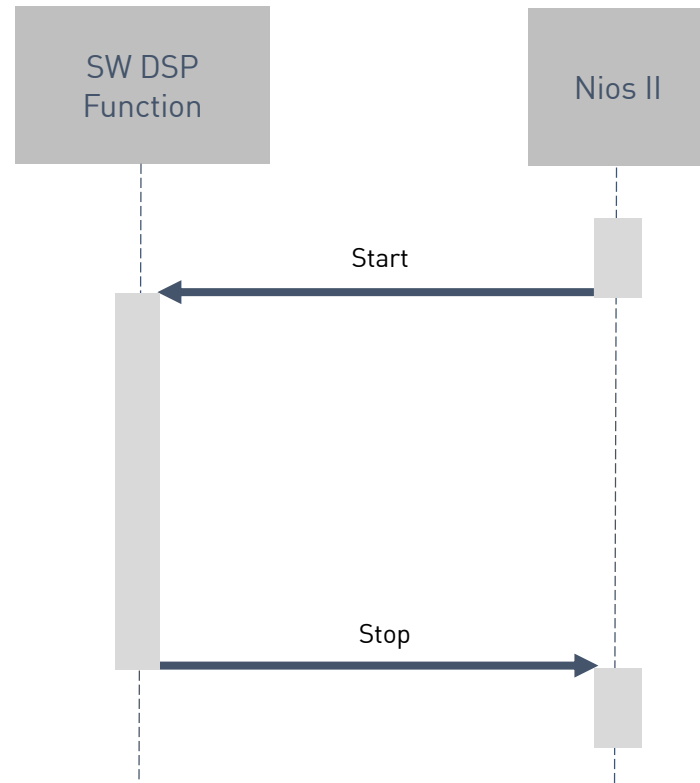
FPGAs in a Nutshell ...

- How to speed up software running on an FPGA internal softcore processor?

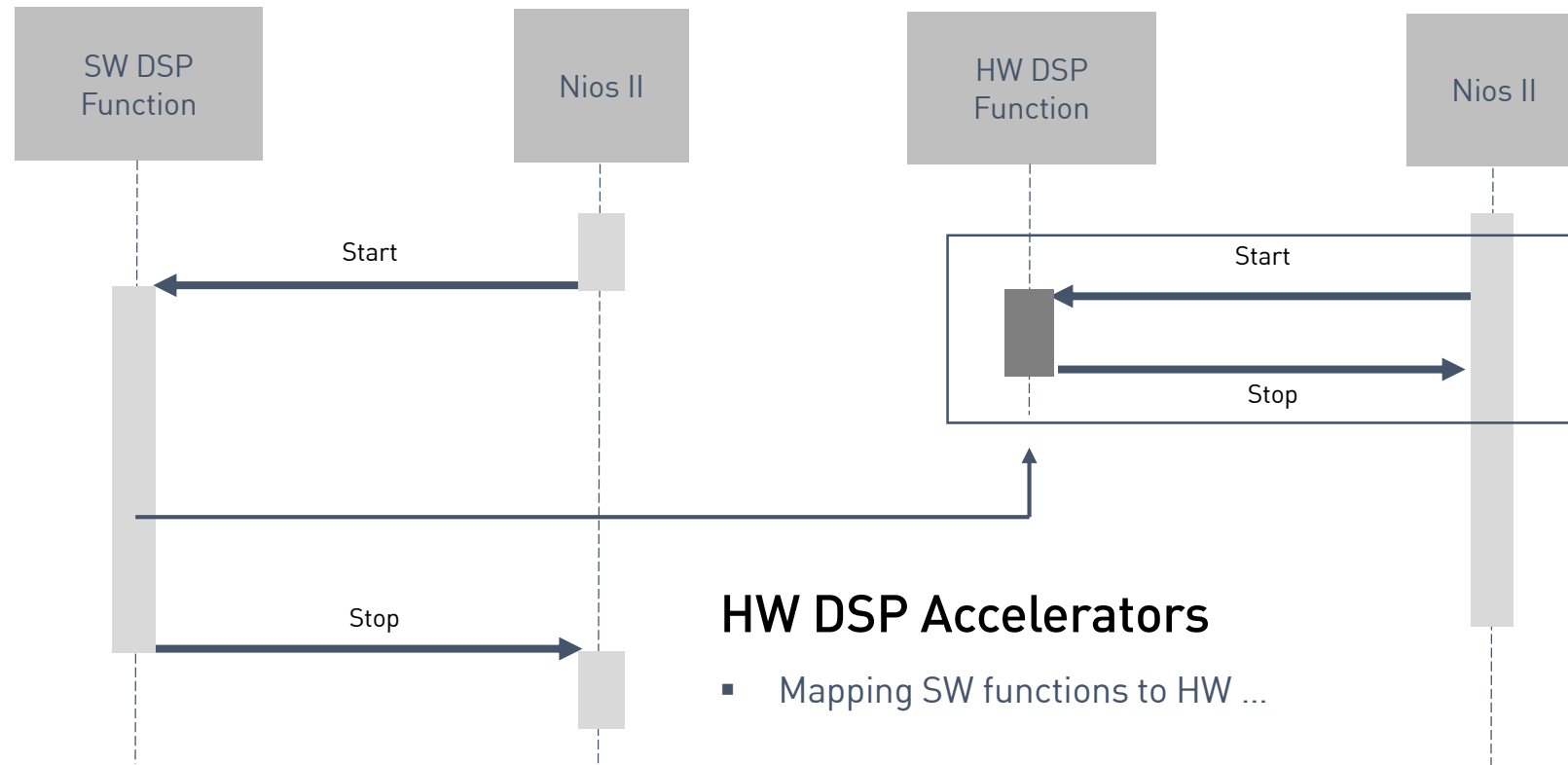
Embedded Signal Processing Systems

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FPGAs in a Nutshell ...



FPGAs in a Nutshell ...



- Parallel execution ...

HW DSP Accelerators

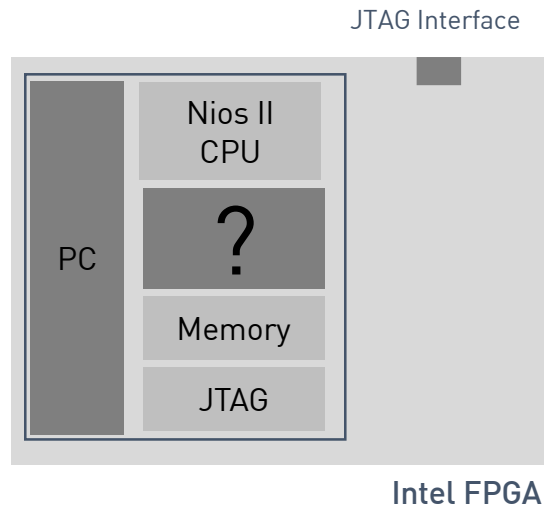
- Mapping SW functions to HW ...

Embedded Signal Processing Systems

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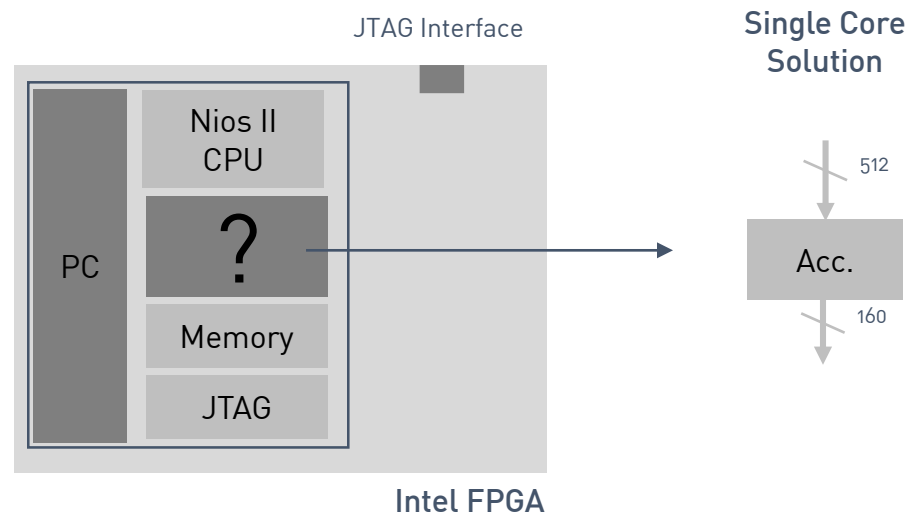
FPGAs in a Nutshell ...

- How to design custom DSP HW Accelerators ...



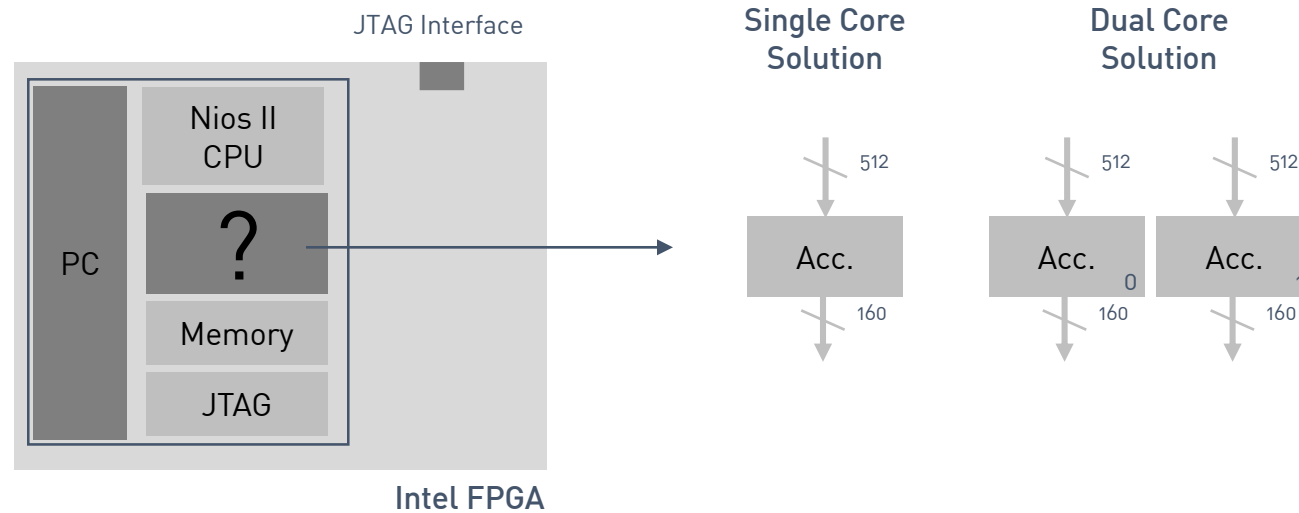
FPGAs in a Nutshell ...

- How to design custom DSP HW Accelerators ...



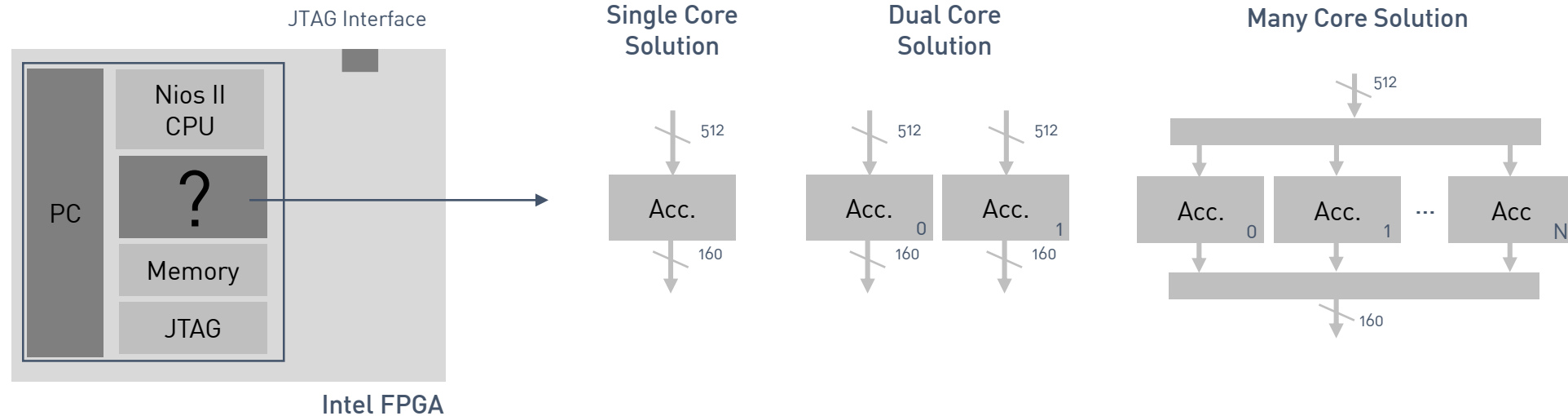
FPGAs in a Nutshell ...

- How to design custom DSP HW Accelerators ...



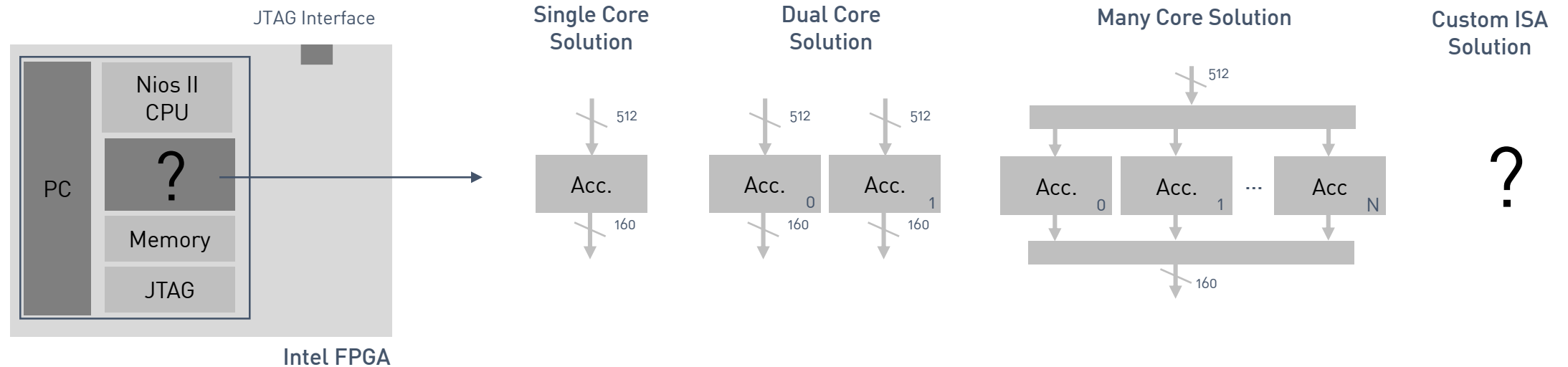
FPGAs in a Nutshell ...

- How to design custom DSP HW Accelerators ...



FPGAs in a Nutshell ...

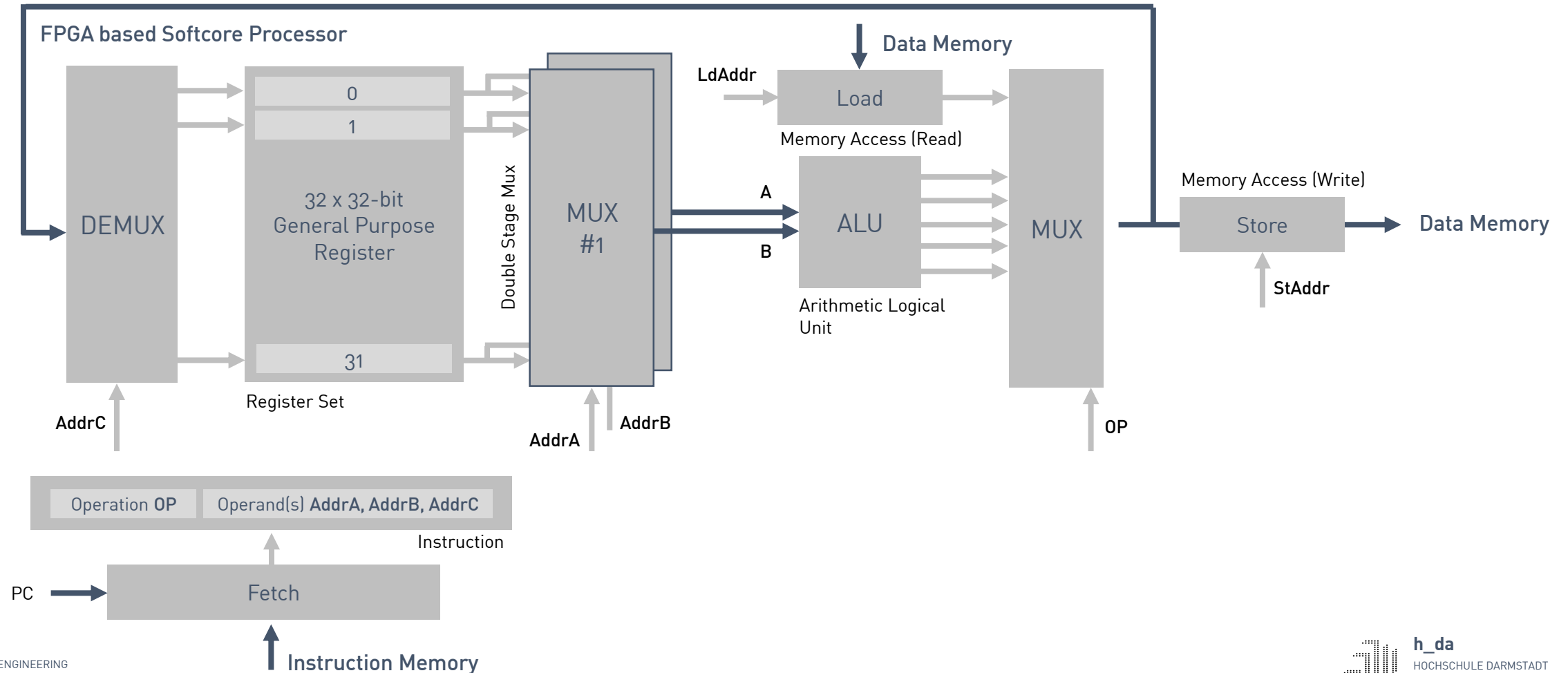
- How to design custom DSP HW Accelerators ...



Embedded Signal Processing Systems

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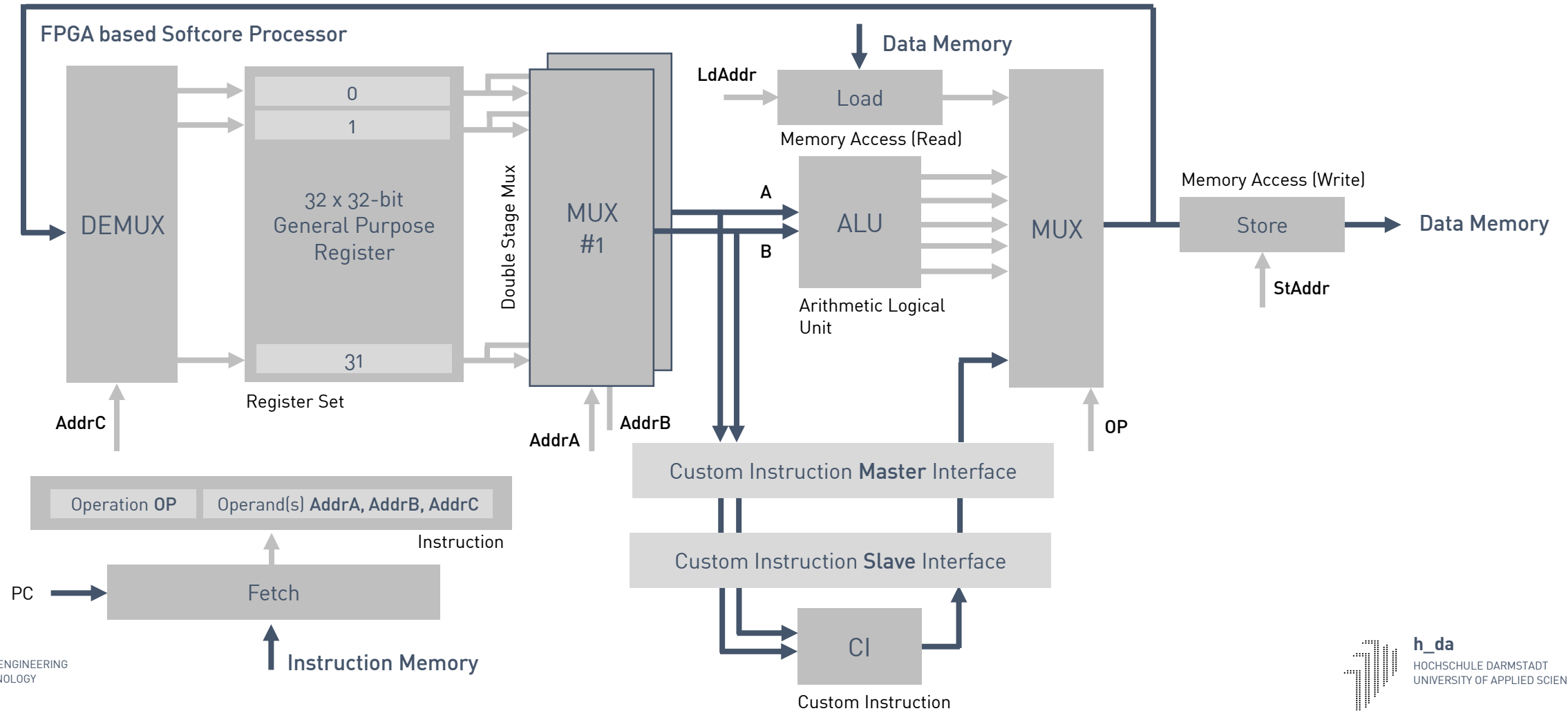
FPGAs in a Nutshell ...



Embedded Signal Processing Systems

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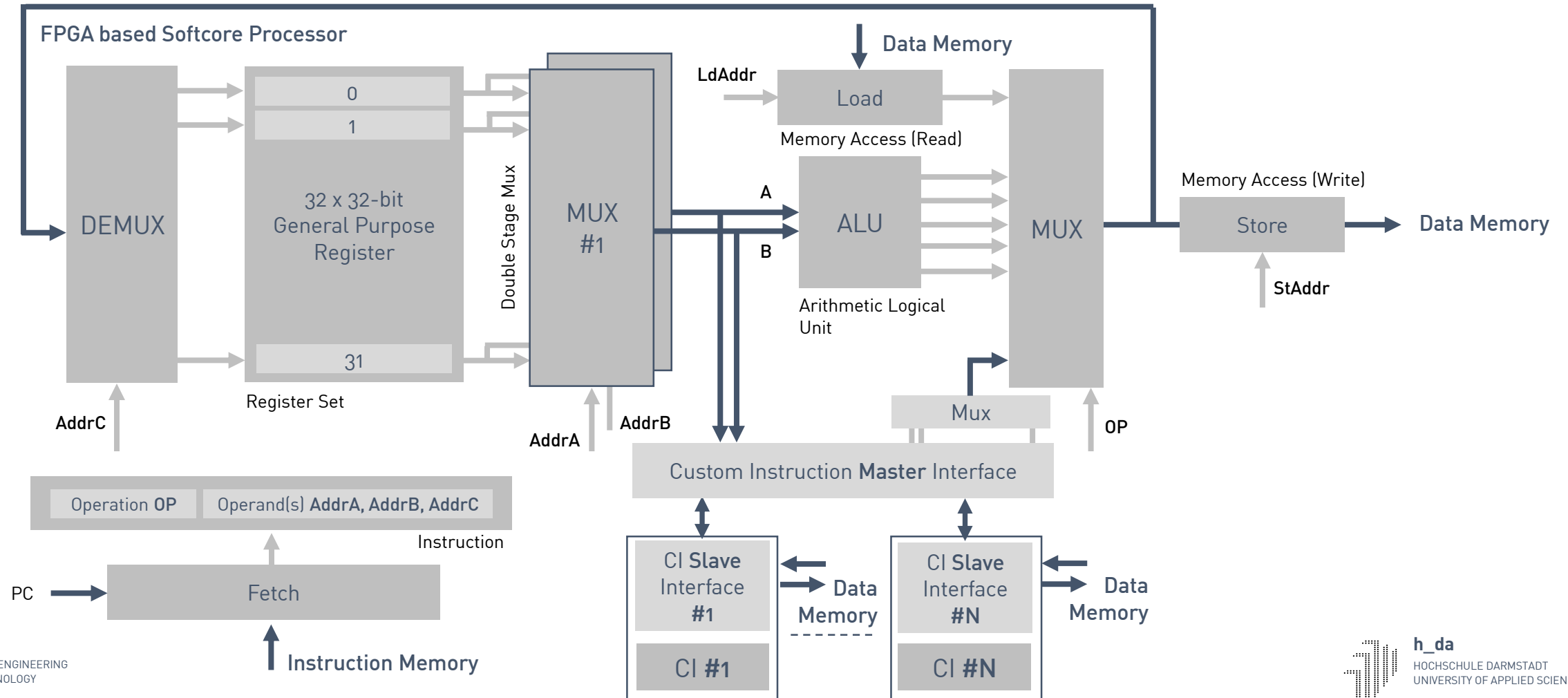
FPGAs in a Nutshell ...



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FPGAs in a Nutshell ...





FPGAs in a Nutshell – Design Space Exploration

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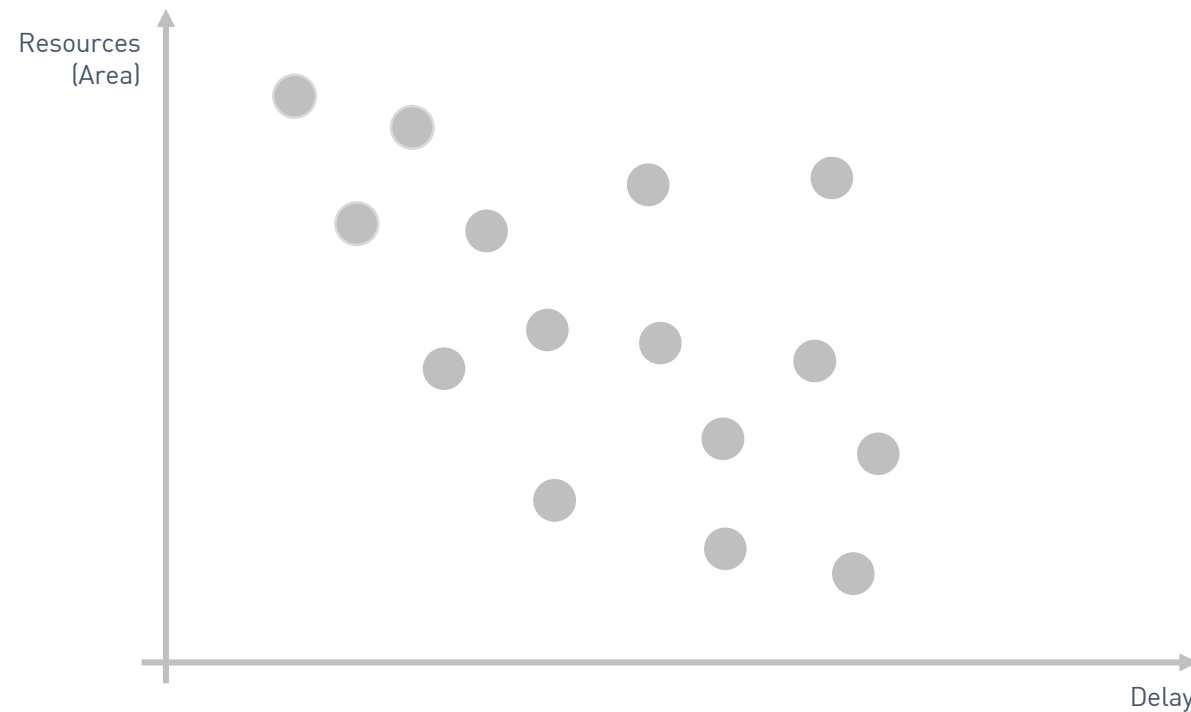
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FPGAs in a Nutshell ...

- FPGA based DSP designs can be characterized by
 - ... the timing of the designed system
 - ... the power consumption of the designed system
 - ... or by the final resource usage

FPGAs in a Nutshell ...

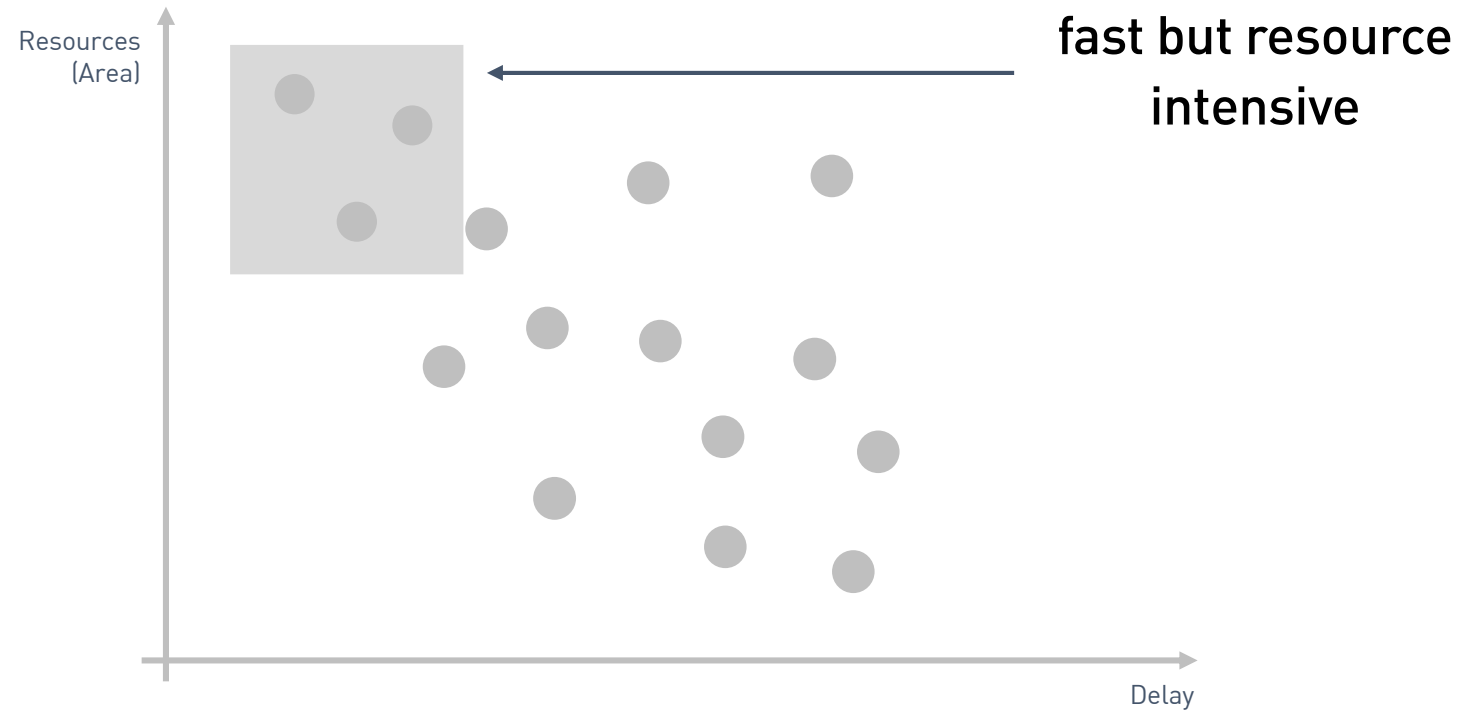


- 15 different designs, each implementing the same functionality
- Which one to choose?

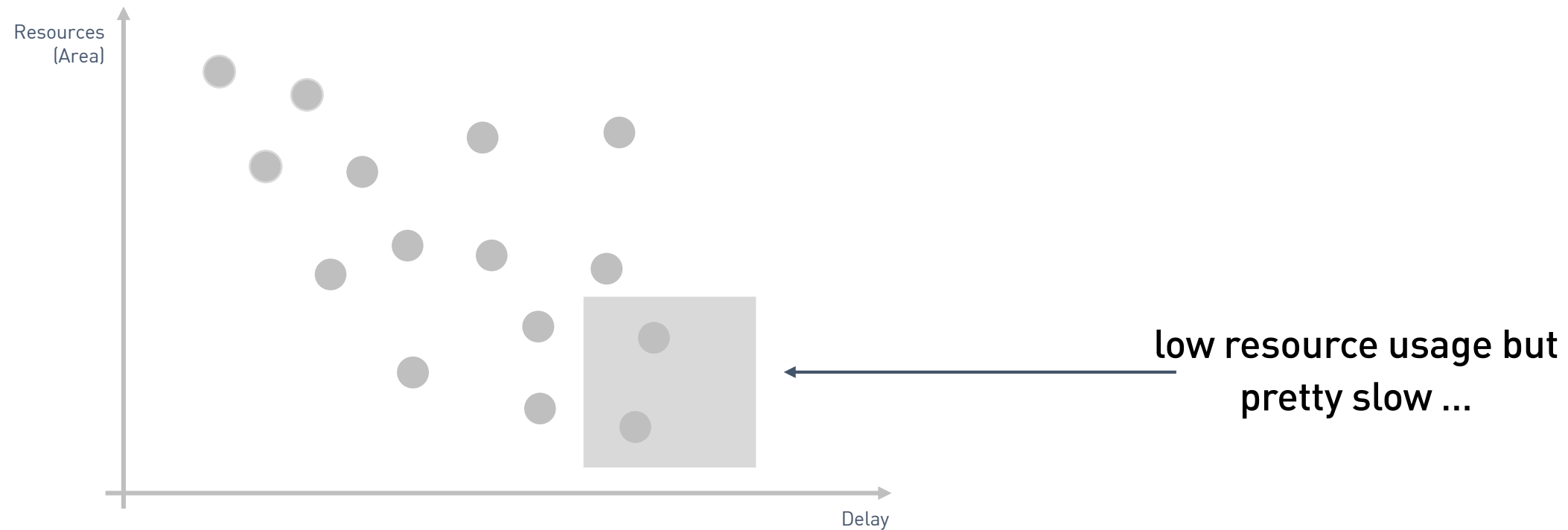
FPGAs in a Nutshell ...

- Which one to use ...?

FPGAs in a Nutshell ...



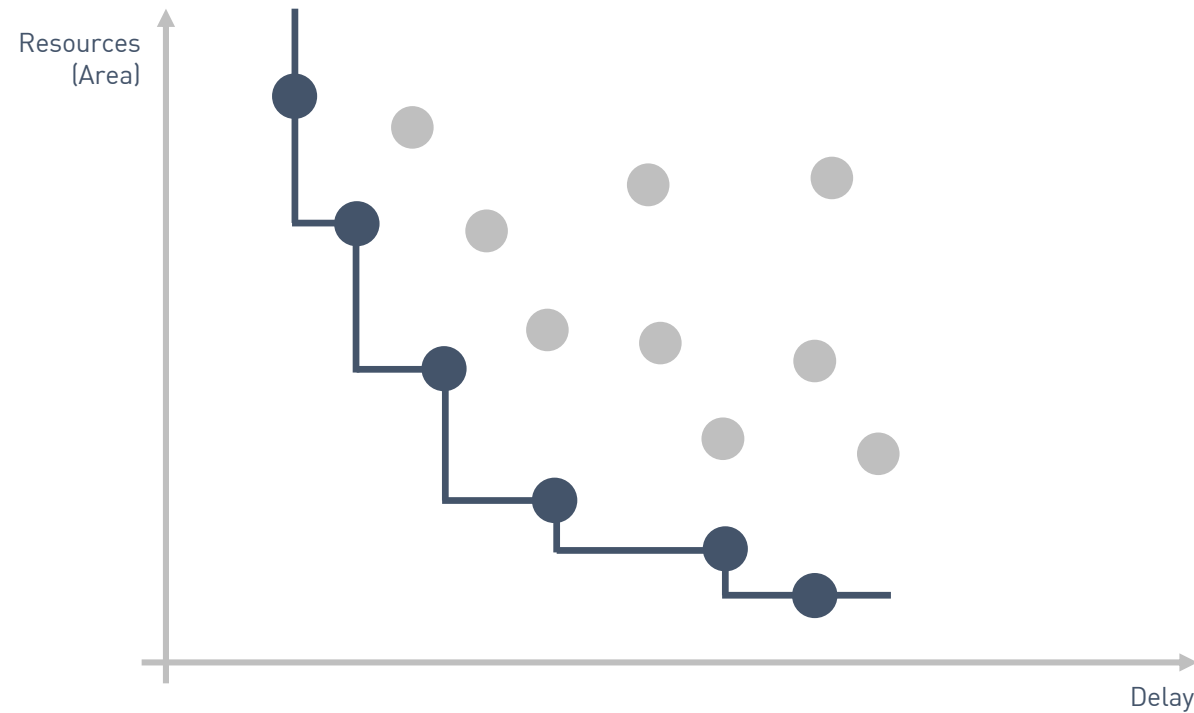
FPGAs in a Nutshell ...



FPGAs in a Nutshell ...

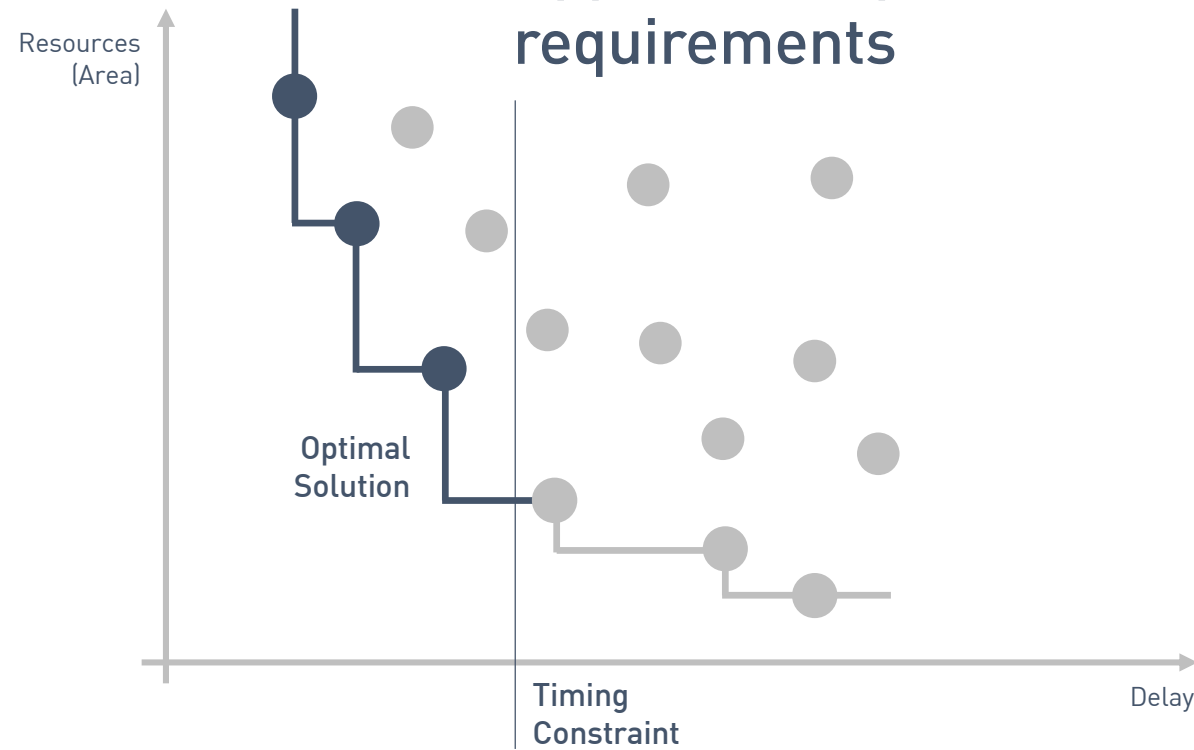
- Again, which one to use ...?

FPGAs in a Nutshell ...

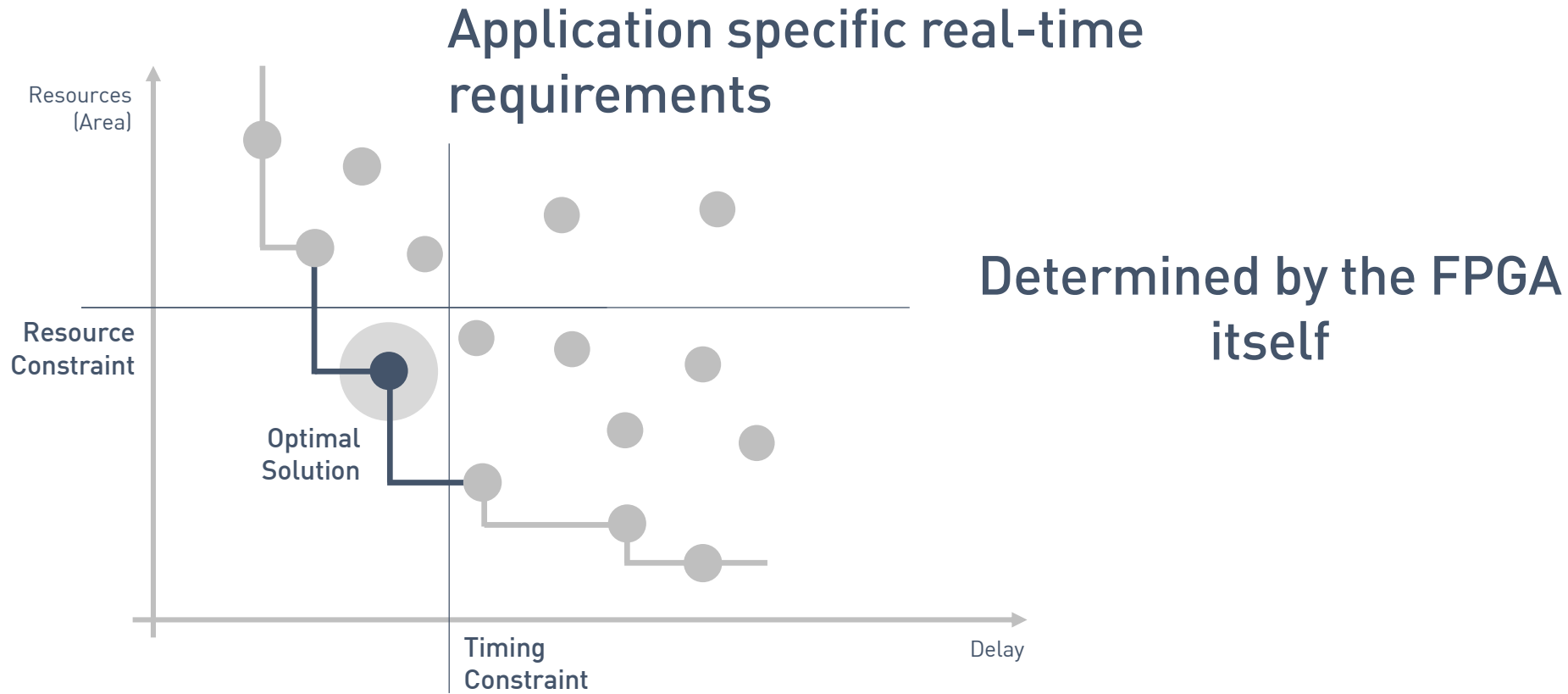


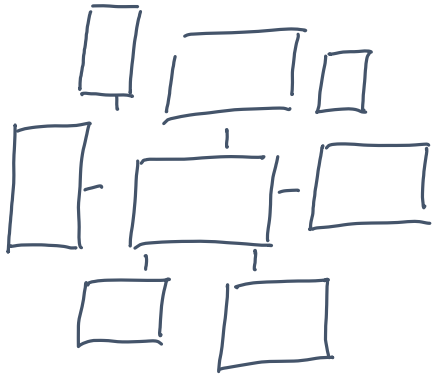
FPGAs in a Nutshell ...

Application specific real-time requirements



FPGAs in a Nutshell ...





Introduction to SystemVerilog HDL – Part #1

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What is an HDL?

- A Language to describe, simulate, and create hardware (popular examples are VHDL, Verilog or SystemVerilog).
- Despite similar syntax, an HDL cannot be used like typical programming languages
- Express the dimensions of timing and concurrency.
- At Register Transfer Level (RTL), an HDL design describes a hardware structure, not an algorithm.
- At behavioral level, HDL models describe only the behavior of the design with no implied structure.
- Something that you should keep in mind: “If you can’t draw it, don’t try to code it!”

How to learn an HDL ...

- Learning by doing.
- Learning from mistakes.
- Try to understand what and why something went wrong ... Otherwise nothing has been learned.
- Start designing simple designs, slowly add complexity.
- Start your design on paper ...

SystemVerilog at a glance ...

- SystemVerilog represents a unified hardware design, specification and verification language.
- It provides support for all Verilog constructs (Verilog-2005). In addition, it combines synthesizable constructs from Accelera's language Superlog and Verification constructs from Synopsys OpenVera.
- In general, the feature-set of SystemVerilog (IEEE standard 1800-2012) can be divided into two distinct sections:
 1. SystemVerilog for RTL design is an extension of Verilog-2005; all features of that language are available in SystemVerilog.
 2. SystemVerilog for verification uses extensive object-oriented programming techniques and is more closely related to Java than Verilog.

SystemVerilog at a glance ...

- It is important to understand that SystemVerilog is both a ‘synthesis’ and ‘simulation/verification’ language.
- A certain subset of the language is used for synthesizing a hardware description into dedicated set of logic gates and flip-flops.
- Another subset of the language provides features for simulation and verification purposes. These constructs can not be translated into equivalent hardware structures ...

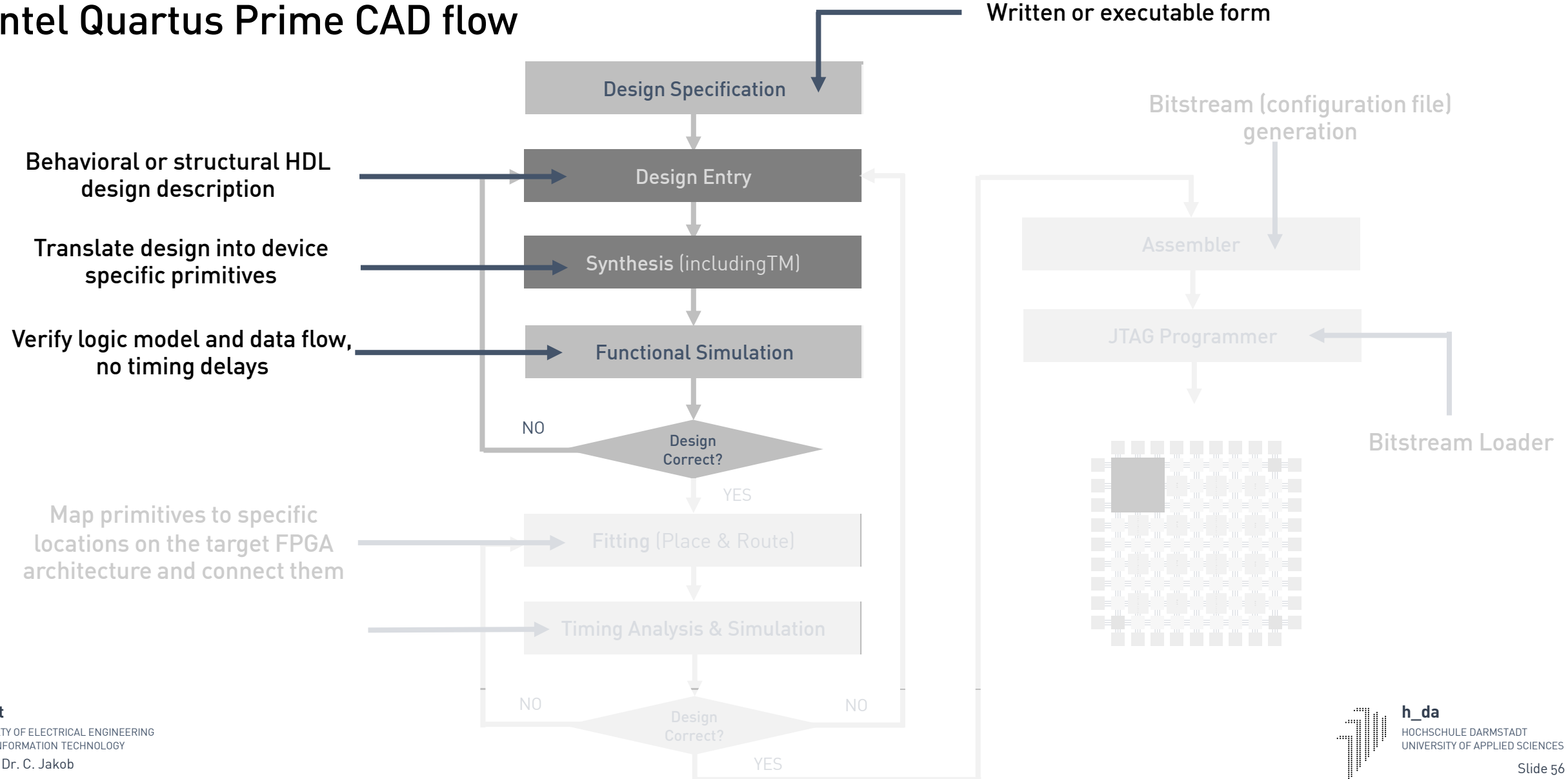
SystemVerilog at a glance ...

- Despite the fact that SystemVerilog syntactically looks like 'C', it is no software programming language.
- This leads to the point that certain construct can be easily misinterpreted.
- It is a good strategy to think of the hardware synthesized that each line of SystemVerilog code will produce.

Introduction to SystemVerilog

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Intel Quartus Prime CAD flow



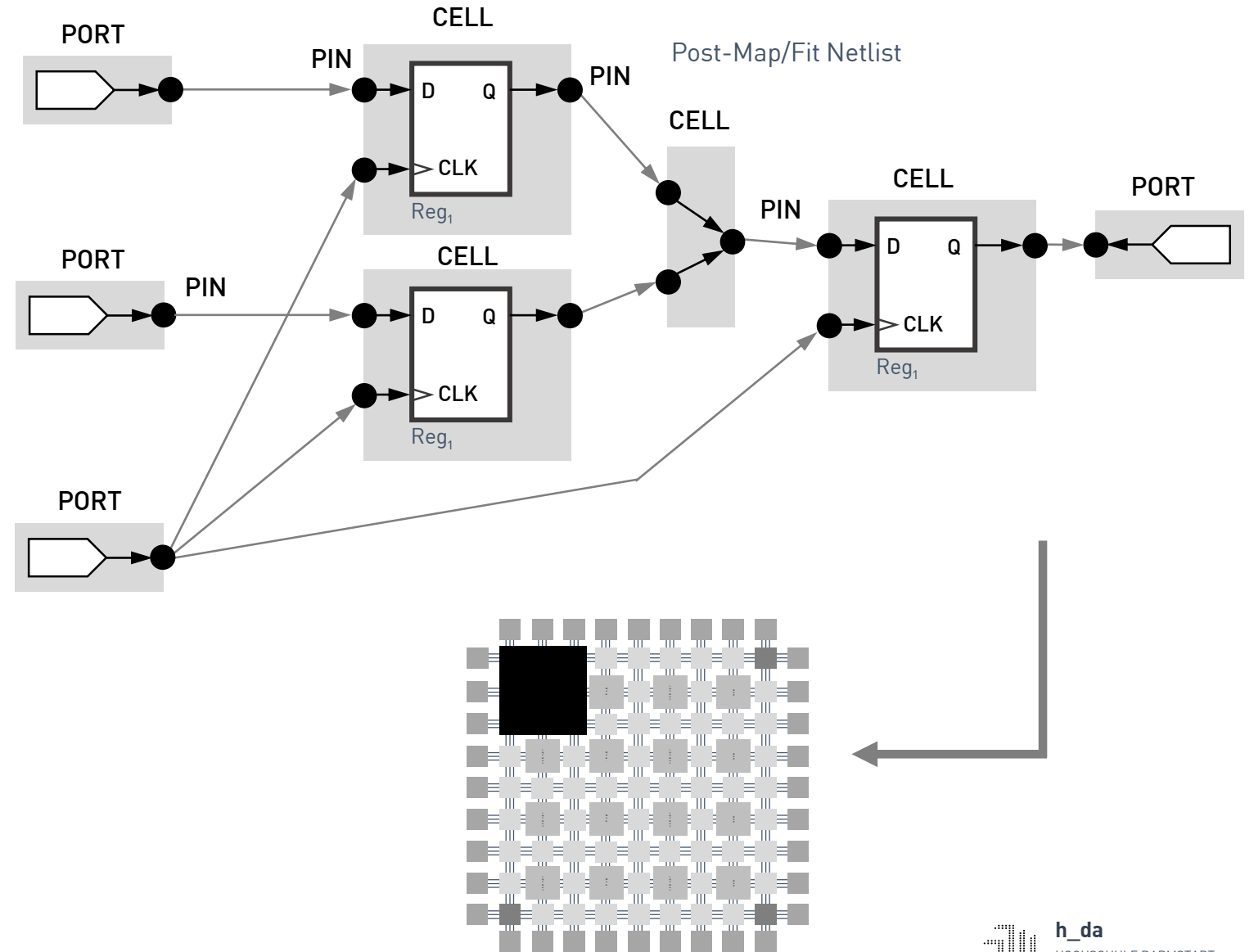
Introduction to SystemVerilog

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What is an HDL?

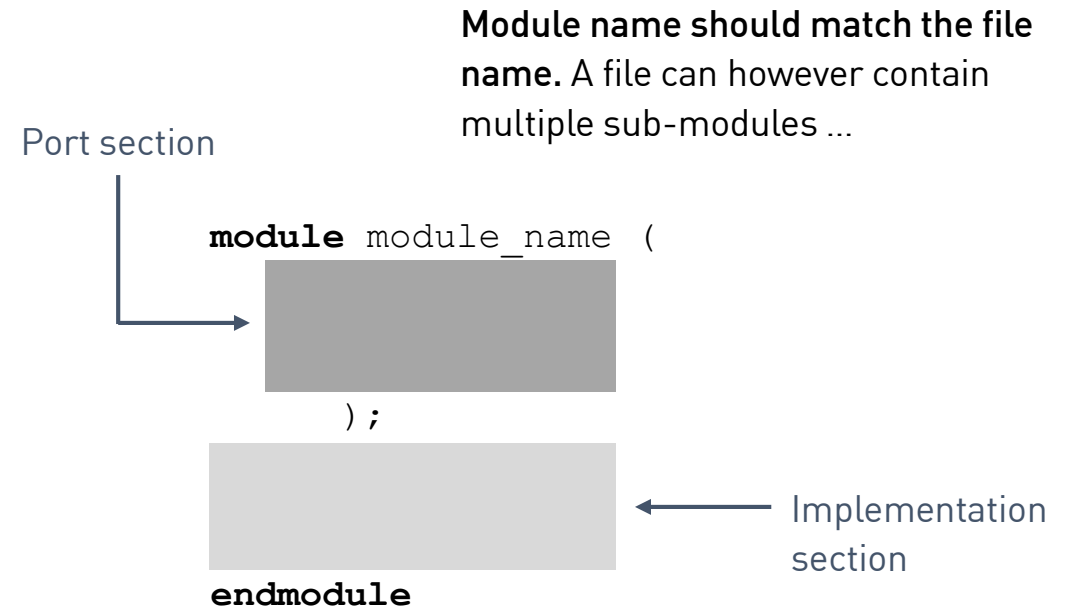
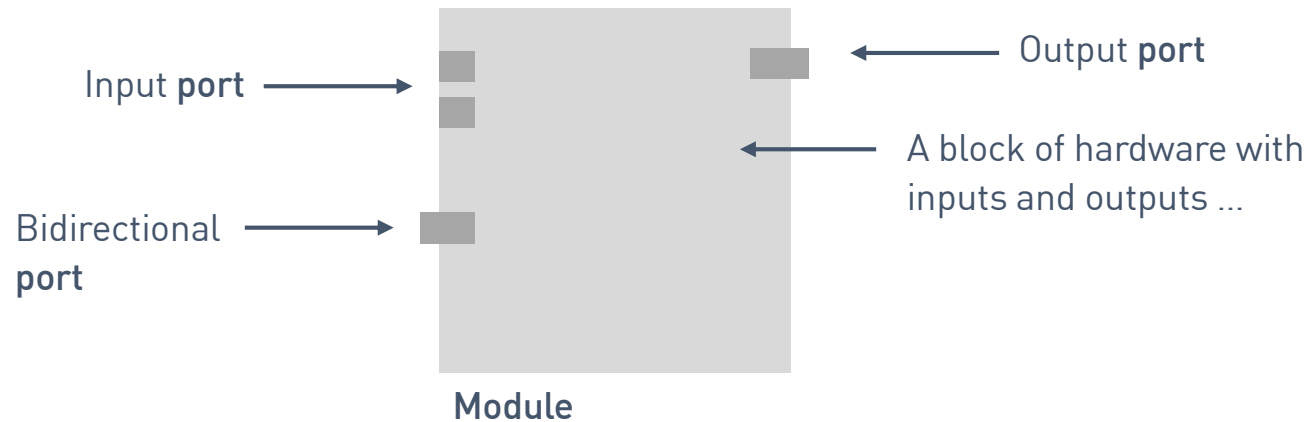
```
1. module sys_proc(  
2.     input logic [3:0] d0, d1,  
3.     ...  
4. );  
5.     ...  
6.     always_ff@(posedge clk)  
7.         if(reset_n == 1'b1)  
8.             q = 1'b0;  
9.         ...  
10. ...  
11. endmodule
```

[SystemVerilog] Source Code Excerpt: sys_proc.sv



The Module – The basic building block in SystemVerilog

- The module is the basic unit of hierarchy in SystemVerilog.
- Modules are used to provide the **coarse-grained** structure of a design.



- The module could implement a simple AND gate, a multiplexer or even something complex such as CPU. In addition, a module can be a single element or collection of lower level modules.
- Ports are used to interface the module with the outside. They are either inputs, outputs or bidirectional (tri-state logic).
- In conclusion, modules describe the boundaries of a design unit [module, endmodule], its inputs and outputs [ports] as well how it works [behavioral or RTL code].
- SystemVerilog is able to model a design at different levels of abstraction. A high level express little detail, low levels express much. Boundaries between levels are often not well defined.

Introduction to SystemVerilog

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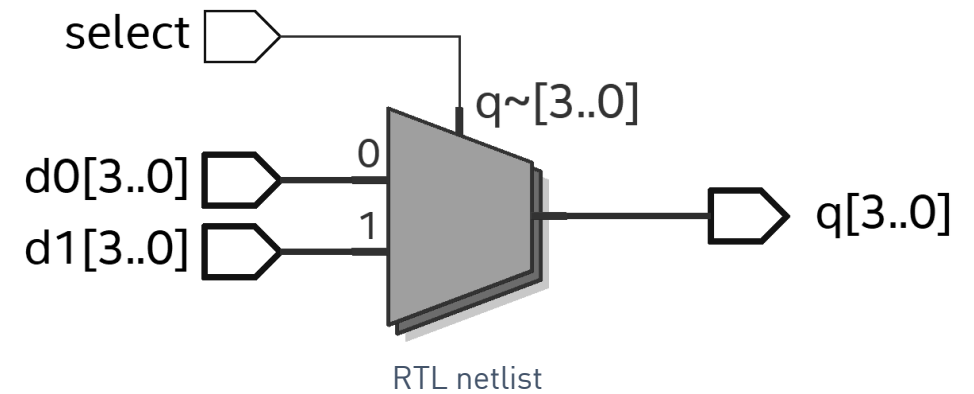
The Module – The basic building block in SystemVerilog

```
1. module mux_2_4bit_comb(  
2.     input logic [3:0] d0, d1,  
3.     input logic select,  
4.     output logic [3:0] q  
5. );  
6.     always_comb  
7.         if(select == 1)  
8.             q = d1;  
9.         else  
10.             q = d0;  
11. endmodule
```

Port or IO
section

Implementation
section

1/1



[SystemVerilog] Source Code: mux_2_4bit_comb.sv

Introduction to SystemVerilog

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The Module – The basic building block in SystemVerilog

Module name, must be identical with the file name

```
1. // ...
2. module full_adder(
3.     input logic d0, d1, cin, output logic q,
4.     output logic cout
5. );
6.
7.     assign q = d0 ^ d1 ^ cin;
8.     assign cout = (d0 & d1) | (d0 & cin) | (d1 & cin);
9.
10. endmodule
```

data typ: single bit

1/1

port name

continuous assignment. another way for describing combinational logic

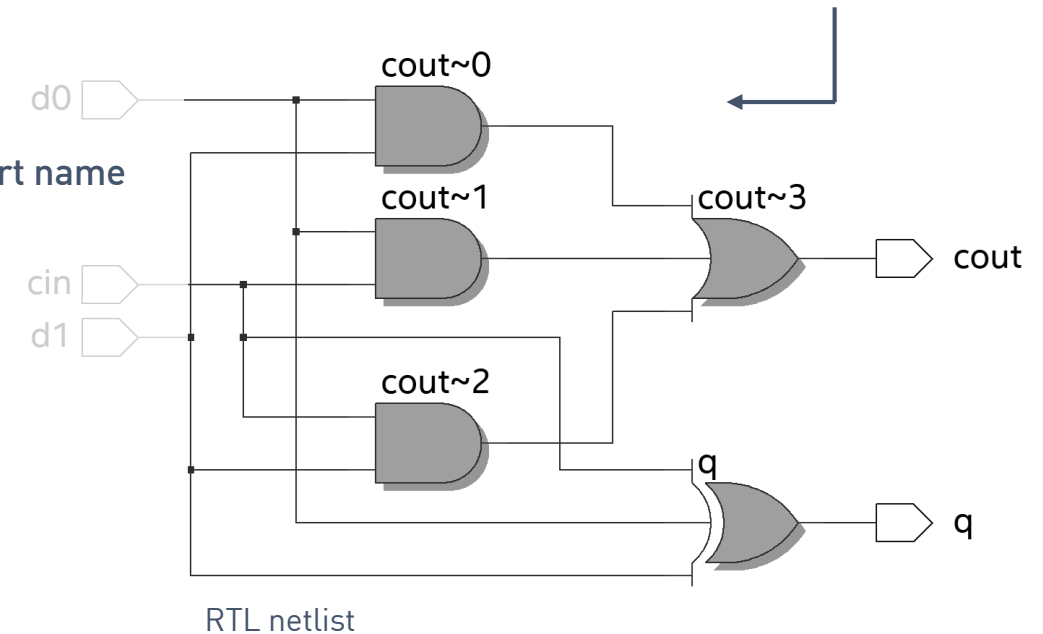
logical expression

[SystemVerilog] Source Code: full_adder.sv

Notes

- SystemVerilog is case sensitive: cin and Cin are not the same
- No names start with numbers

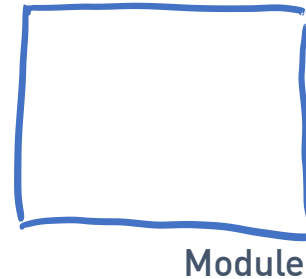
This is a **graphical representation** of the design netlist after Analysis & Elaboration and netlist extraction, but before Quartus Prime synthesis and fitting optimizations. **This RTL netlist is not the final structure of the design**, because not all optimizations are included; instead it is the closest possible view to the original RTL design



Structural and behavioral HDL descriptions

```
module full_adder(  
    input logic d0, d1, cin,  
    output logic q, cout  
);  
  
    logic tmp_sum, tmp_cout_0, tmp_cout_1;  
  
    half_adder inst_0( .d0(d0), d1(d1), .q(tmp_sum),  
                      .cout(tmp_cout_0));  
    half_adder inst_1( .d0(cin), d1(tmp_sum), .q(q),  
                      .cout(tmp_cout_1));  
  
    assign cout = tmp_cout_0 | tmp_cout_1;  
  
endmodule
```

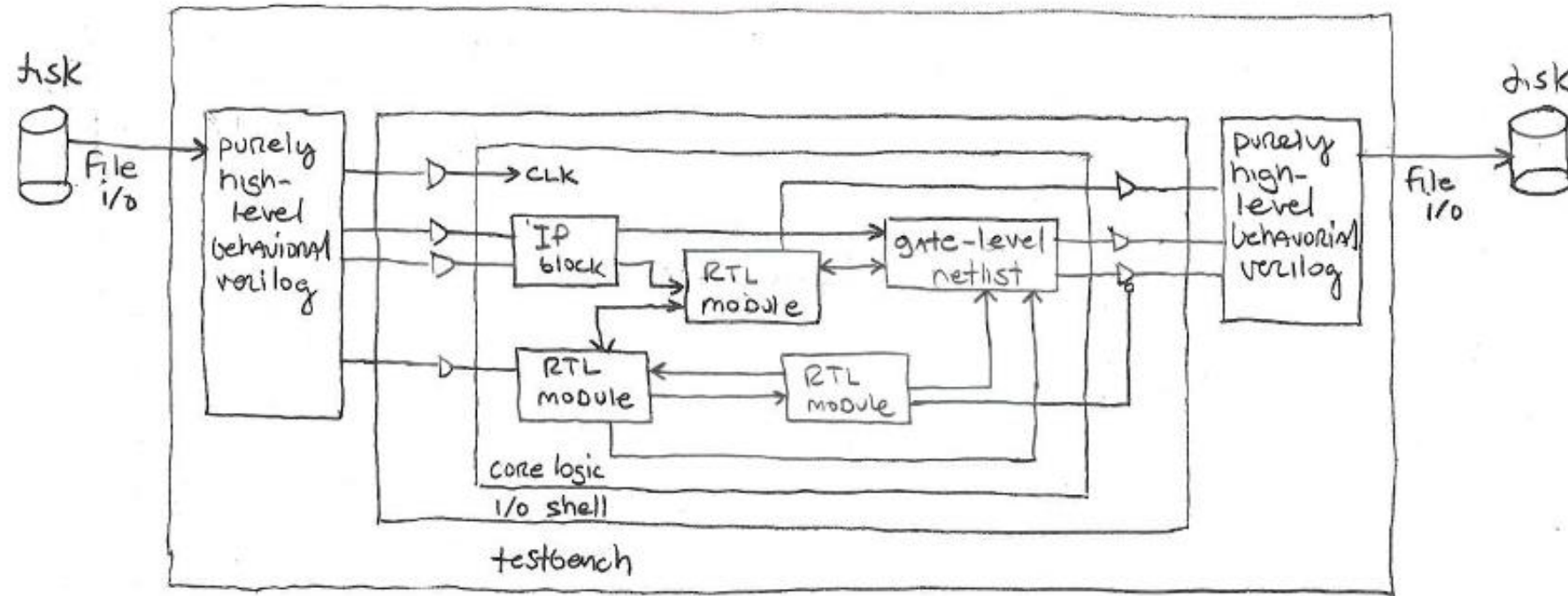
Structural description, describes
the interconnect and usage of
lower-level components



```
module add_sub_4bit(  
    input logic [3:0] d0, d1, input logic cin,  
    output logic [3:0] q, output logic cout  
);  
    function logic [4:0] add (input logic [3:0] a, b,  
    input logic cin );  
        logic [3:0] s; logic c; c = cin;  
        for(int i = 0; i < 4; i++) begin  
            s[i] = a[i] ^ b[i] ^ c; c = (a[i] & b[i]) | (a[i] &  
                | (c & c[i]));  
        end  
        adder = {c, s};  
    endfunction  
  
    always_comb  
        if(operation)  
            {cout, q} = adder(d0, ~d1, 1);  
        else  
            {cout, q} = adder(d0, d1, 0);  
  
endmodule
```

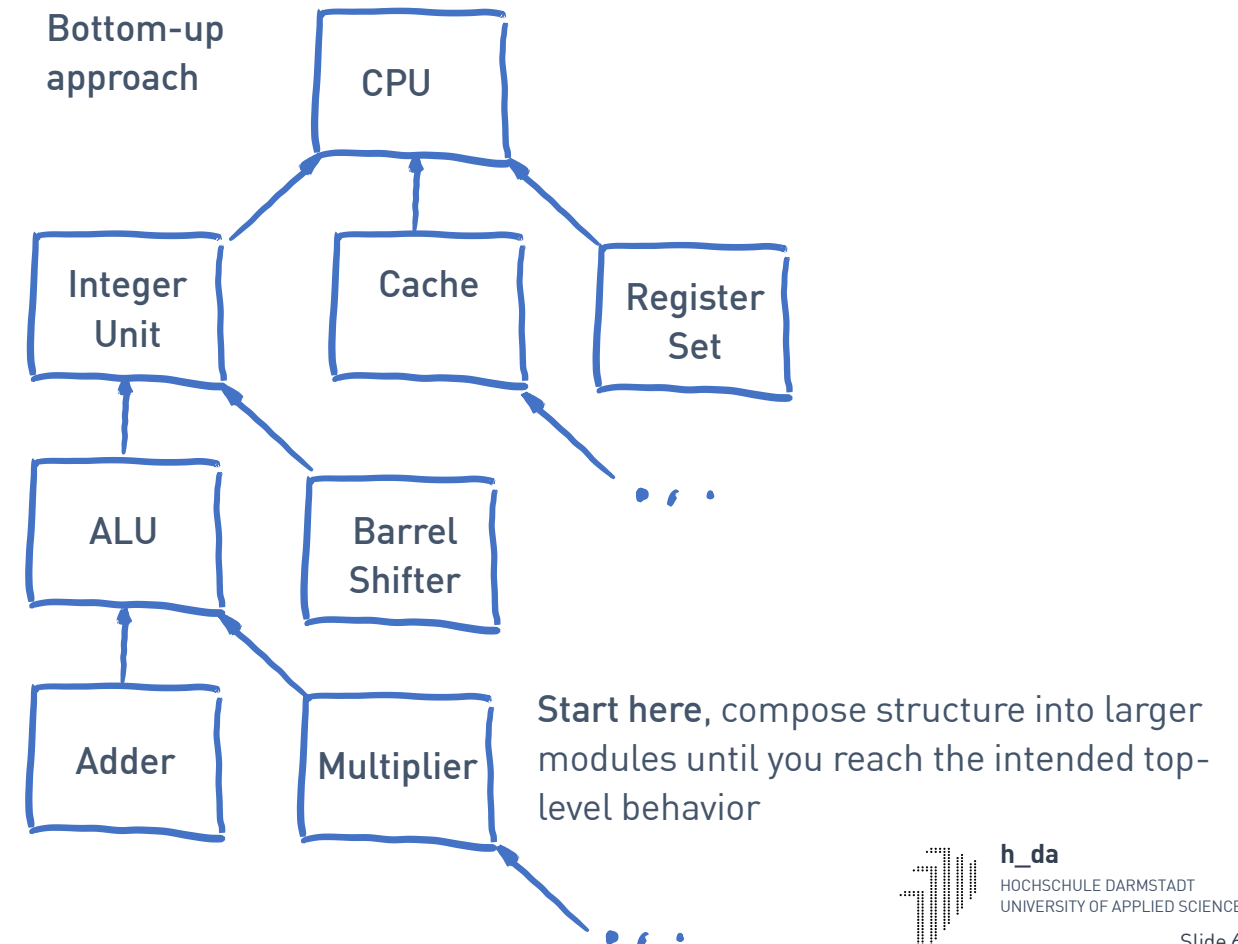
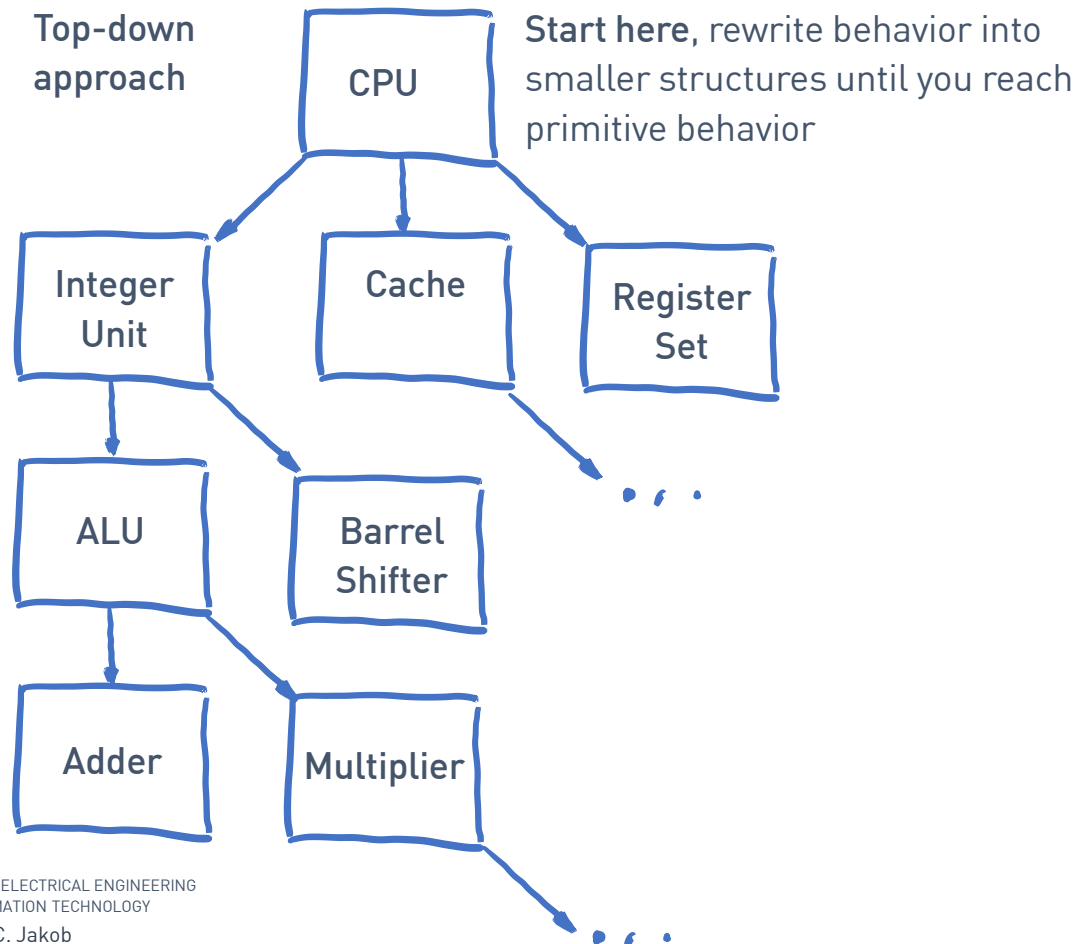
Behavioral description, describes
the algorithmic behavior of the
module rather than its structure

The Module – The basic building block in SystemVerilog



- A hierarchical design has a top level module and lower level ones. Lower level modules are instantiated within the higher level module. Lower level modules are connected together with wires.

Top-down and bottom-up design HDL descriptions



Introduction to SystemVerilog

h_da – fbeit - Embedded Signal Processing Systems

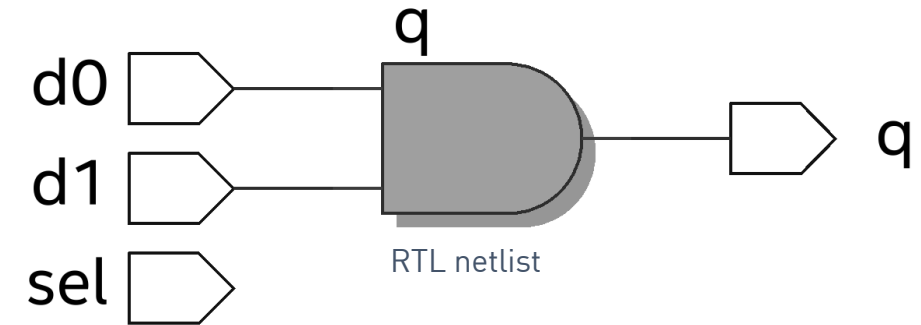
Describing Combinational Logic using SystemVerilog

A series of basic SystemVerilog designs

Simple AND gate - Logical bit-wise operators

```
1. module and_2_1bit(  
2.     input logic d0, d1,  
3.     output logic q  
4. );  
5.  
6.     // 2-input AND gate  
7.     assign q = d0 & d1;  
8.  
9. endmodule
```

1/1



[SystemVerilog] Source Code: and_2_1bit.sv

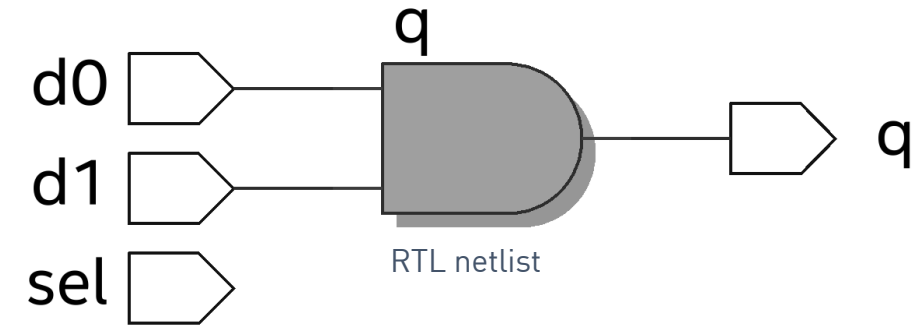
Notes

- The above example shows a so called continuous assignments (**assign**). These constructs are intended for modelling combinational logic.
- A continuous assignment drives a net similar to how a gate drives a net. The expression on the right hand side can be thought of as a combinatorial circuit that drives the net continuously.

Simple AND gate - Logical bit-wise operators

```
1. module and_2_1bit(  
2.     input logic d0, d1,  
3.     output logic q  
4. );  
5.  
6.     // 2-input AND gate  
7.     always_comb  
8.         q = d0 & d1;  
9.  
10. endmodule
```

1/1



[SystemVerilog] Source Code: and_2_1bit.sv

Notes

- An alternative way to model combinational logic is to use the `always_comb` construct (in conjunction with **blocking** assignment operators).
- The `always_comb` implicitly creates a complete sensitivity list including all variables and nets that are read in the process (pretty much the same always as `@*` construct in Verilog-2001).
- Please note that variables on the LHS of assignments cannot be assigned by other procedural blocks.

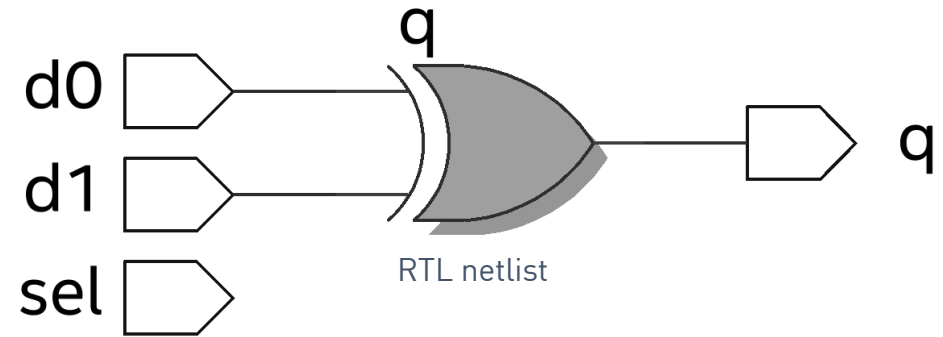
Introduction to SystemVerilog

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Simple XOR gate - Logical bit-wise operators

```
1. module xor_2_1bit(  
2.     input logic d0, d1,  
3.     output logic q  
4. );  
5.  
6.     // 2-input XOR gate  
7.     assign q = d0 ^ d1;  
8.  
9. endmodule
```

1/1



[SystemVerilog] Source Code: xor_2_1bit.sv

Notes

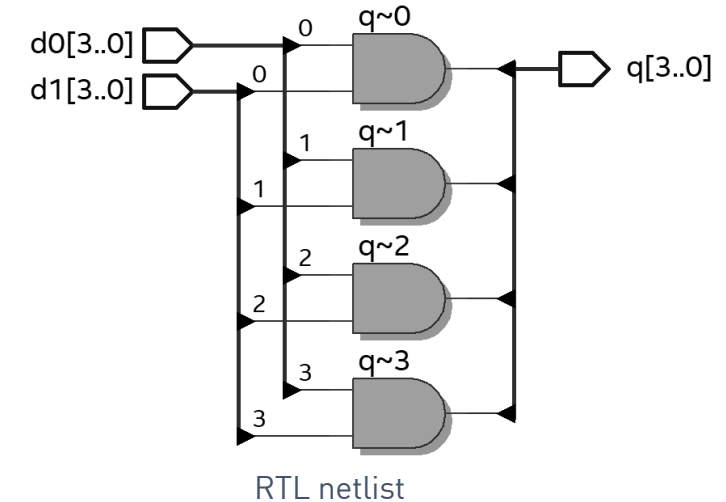
- Bit-wise AND: &
- Bit-wise OR: |
- Bit-wise XOR: ^
- Bit-wise NOT: ~

Simple AND gate - Logical bit-wise operators

```
1. module and_2_4bit(  
2.     input logic [3:0] d0, d1,  
3.     output logic [3:0] q  
4. );  
5.  
6.     // four 2-input AND gates  
7.     assign q = d0 & d1;  
8.  
9. endmodule
```

1/1

- Four bit wide vector, little-endian convention



[SystemVerilog] Source Code: and_2_4bit.sv

Notes

- Bitwise operators perform bit-oriented operations on vectors

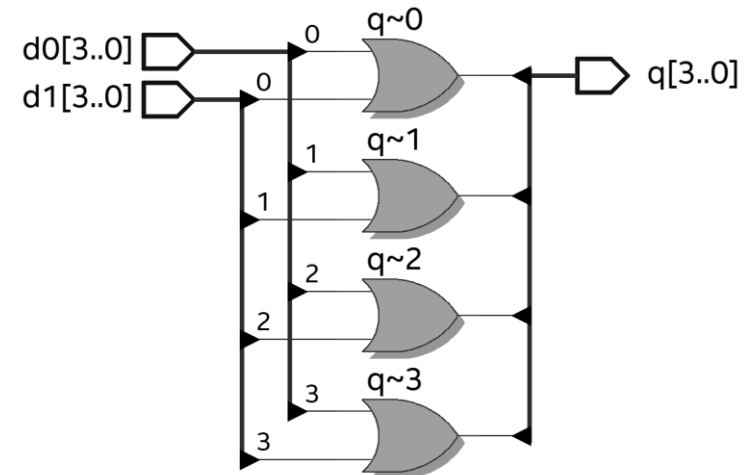


Multi-bit Boolean gates - Logical bit-wise operators

```
1. module or_2_4bit(  
2.     input logic [3:0] d0, d1,  
3.     output logic [3:0] q  
4. );  
5.  
6.     // four 2-input OR gates  
7.     assign q = d0 | d1;  
8.  
9. endmodule
```

1/1

[SystemVerilog] Source Code: or_2_4bit.sv



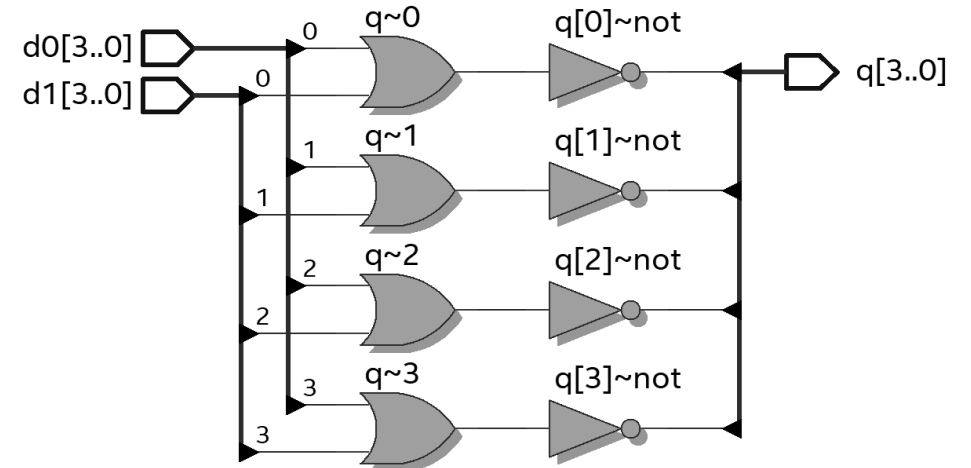
RTL netlist

Multi-bit Boolean gates - Logical bit-wise operators

```
1. module nor_2_4bit(  
2.     input logic [3:0] d0, d1,  
3.     output logic [3:0] q  
4. );  
5.  
6.     // four 2-input NOR gates  
7.     assign q = ~(d0 | d1);  
8.  
9. endmodule
```

1/1

[SystemVerilog] Source Code: nor_2_4bit.sv



Logical operators

```
1. module logical_and_2_4bit(  
2.     input logic [3:0] d0, d1,  
3.     output logic q  
4. );  
5.  
6.     assign q = (d0 && d1);  
7.  
8. endmodule
```

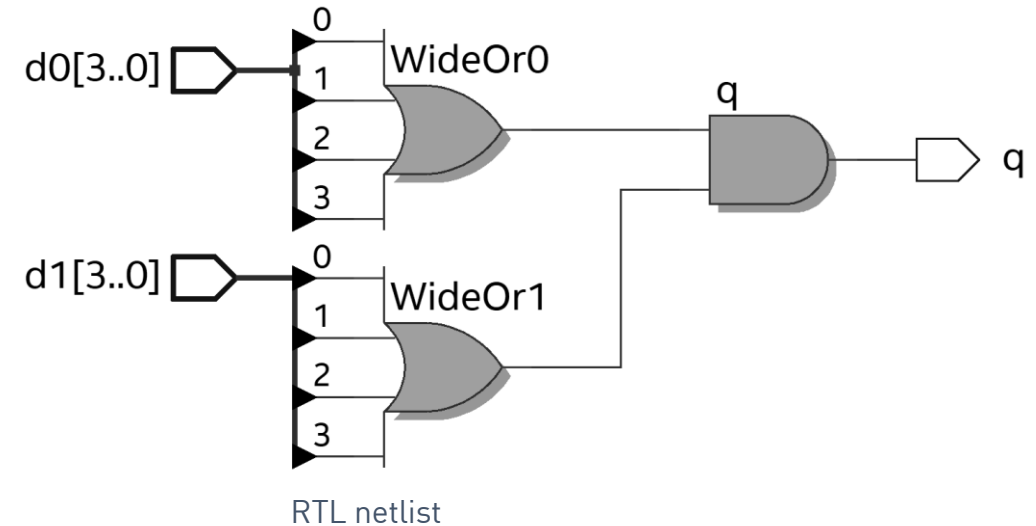
1/1

[SystemVerilog] Source Code: logical_and_2_4bit.sv

Notes

- Logical operators evaluate to a 1-bit value.
- Operands not equal to zero are equivalent to one.
- Verilog Logical Operators:

Logical AND:	&&
Logical OR:	
Logical NOT:	!





Appendix

International Master of Science in Electrical Engineering

Prof. Dr. C. Jakob

University of Applied Sciences Darmstadt

h_da

Faculty of Electrical Engineering and Information Technology

fbeit

History of SystemVerilog

- Verilog was developed by Gateway Design Automation as a proprietary language for logic simulation (Verilog-XL simulator) in 1984. The term Verilog is made-up of the two words "verification" and "logic".
- Gateway Design Automation was acquired by Cadence Design Systems in 1989.
- Verilog became an open standard in 1990 when Cadence Design Systems released Verilog to the public domain. The administration of the language was assigned to the Open Verilog International (OVI) organization which was founded in the same year. OVI had the task of taking the language through the IEEE standardization process.
- In 1995, Verilog was adopted as an IEEE standard (IEEE standard 1364-1995, also denoted as Verilog-95).
- A significantly revised version of Verilog-95 became an IEEE standard in 2001 (IEEE standard 1364-2001, also known as Verilog-2001).

History of SystemVerilog contd.

- The Accellera Systems Initiative was founded in 2000 through the unification of Open Verilog International and VHDL International. Accellera was established as an independent, non-profit organization dedicated to create, support, promote, and advance system-level design, modeling, and verification standards.
- In the early 2000s, Accellera started to develop SystemVerilog as an extension of the Verilog IEEE 1364-2001 standard.
- In 2005, SystemVerilog was adopted as an IEEE standard (1800-2005).
- In the same year, a (slightly) modified and extended Version of Verilog-2001 became an IEEE Standard (IEEE 1364-2005).
- In 2009, the SystemVerilog standard (IEEE 1800-2005) was merged with the base Verilog (IEEE 1364-2005) standard, forming IEEE Standard 1800-2009.
- The current version is IEEE standard 1800-2012.